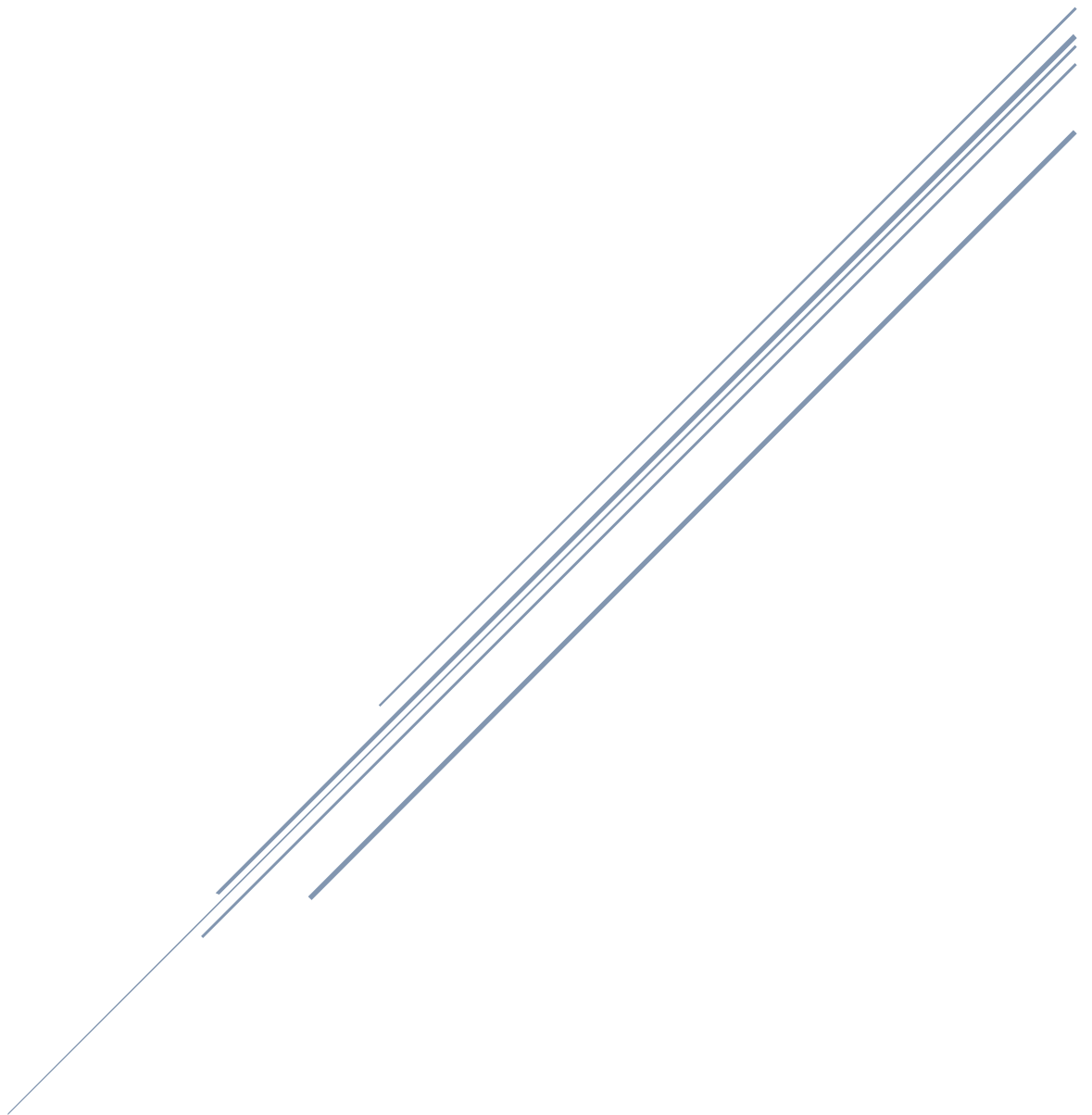


ENHANCING SUPPLY CHAIN INFORMATION
SHARING SYSTEMS THROUGH BLOCKCHAIN-
INTEGRATED CUSTOMER REVIEWS LLM
ANALYSIS: A SMART CONTRACT APPROACH
Research Project Design Practice and Ethics



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Acknowledgments

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Enhancing Supply Chain Information Sharing Systems through Blockchain-Integrated Customer Reviews LLM Analysis: A Smart Contract Approach

Abstract

The research explores the ways in which blockchain technology and AI sentiment analysis can be integrated into the management of complex supply chains to make it not only more transparent and efficient but indeed responsive. In this regard, a number of challenges were underscored as constituent features of the traditional models of the supply chain, including fragmentation in data and lack of real-time information sharing. The research, based on the decentralized and tamper-proof ledger mechanism of blockchain, goes on to propose solutions for improving information-sharing practices across supply chains at all levels, from supplier to consumer. Through a review of the related literature in detail, this study examines the current studies related to the uses of blockchains in supply chains and develops a conceptual framework that considers scalability, regulatory issues, and interoperability with existing systems. These empirical case studies give visibility to how blockchain is applied in practice, thus validating the framework and giving evidence-based recommendations for industry practitioners. The study also identifies best practices and guidelines for successful implementation in various industries with different configurations of blockchain in their supply chains. The findings have brought to light the potential of blockchain to transform supply chain management for greater transparency, reduced inefficiencies, and enhanced customer satisfaction, hence contributing significantly to both knowledge and industry practice.

1. Introduction

1.1. Introduction

The thesis introduction chapter showcased the extensive study of the implementation of Blockchain and multiple Artificial Intelligence (AI) models into supply chain network. It breaks down the underlying need to comprehend and assemble the technologies potential uses in an industry that are full of fraud and mistrust. The introduction chapter put pressure on the importance of applying such technologies into the existing supply chain system to enhance the trackability, traceability, and rapid customer review feedback through the explanation of the context and background. The purpose of the study is to bridge the gap of the real-world supply chain problem with the implementation of the technological improvements. Upon deeper exploration of the dissertation, the readers are guaranteed to possess a sufficient understanding of the research importance, goals, and the wider result consequences.

1.2 Background

The technology rapid growth, especially in the field of Blockchain and LLM Artificial Intelligence, had invented diverse use cases and advancement in various industries by automating and eliminating processes that are usually done by human. Traditional supply chain system is plagued with multiple issues, like lack of trackability and traceability, and huge communication gap between the industry and the customer, resulting in growing unsettled complaints. Hence, it is imperative to comprehend the benefit of both technologies, such problems can be solved if the implementation can be done correctly by addressing the issues properly.

1.3. Problem Definition

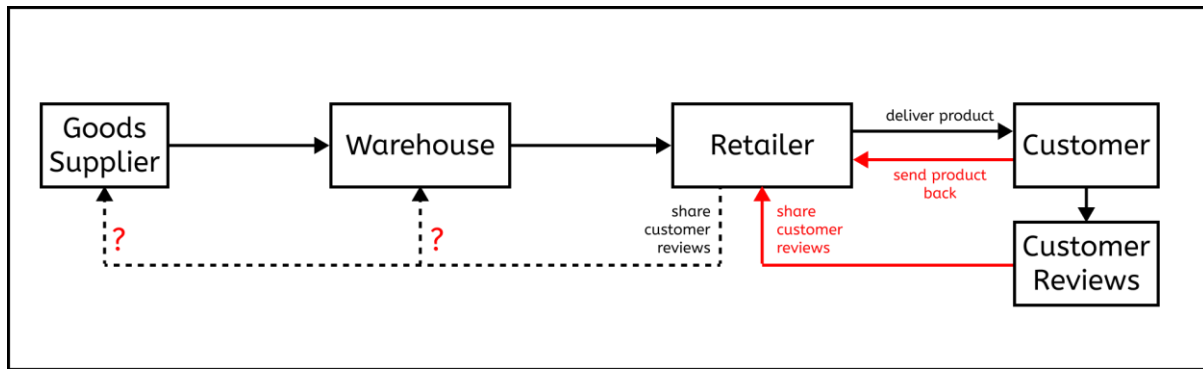


Figure 1. Supply Chain Customer Feedback Sharing Gaps

To explore and elaborate on the research problem stated earlier regarding the complex supply chain dynamics, the investigation must systematically examine several interconnected components that contribute to customer satisfaction and inventory management. This entails a detailed examination of the supply chain's capacity to meet consumer expectations in terms of product availability, quality, and delivery standards, and how discrepancies in these areas lead to customer dissatisfaction and returns.

The basic supply chain network, consisting only goods suppliers, warehouses, retailers, and customers. Each step is conducted semi-independently, resulted in informational silos. The lack of systematic process that integrate suppliers and warehouses' operational decisions and customer feedback leads to a critical inefficiency. This disintegration affected the feedback loop speed, effectivity, or simple non-existent, hence, impeded fast consumer needs and quality feedback adaptation ability.

Challenges like operational challenges and customer impact are the two factors that holding back customer feedback loop within the supply chain network. Firstly, customer review delays are plaguing the supply chain, causing product offering and customer expectations misalignment, which resulted in the operational inefficiencies like in-demand items understocking and unpopular products overstocking. Secondly, the problems being faced by the customers such as product quality or suitability, lead to exchanges and returns high rates. The problems have direct impact to the customer loyalty and satisfaction due to the current system is not effectively provide customer insights to conduct customer service and product improvement.

Due to the previously mentioned problems, supply chain organisations must face several issues such as higher operational costs and the highly competitive market due to the existence of new sophisticated digital tools. To begin with, due to the inefficiencies of resources management to handle product returns and stock management that inaccurately serve customer needs leads to the costlier operation costs. Moreover, important data that can improve product quality and operational efficiency is underutilised across the industry. Furthermore, the enhancement in the technology sector has bring forth a lot of improvement to the supply chain environment in such a rapid pace, thus, moving further the competitiveness of the industry, where significant insight can be extracted easily through the digital tools. A more efficient customer data utilisation competitors threaten the overall

market share, thus, real-time customer feedback integration to the operation of supply chain is both beneficial and necessary.

In the current fiercely competitive industry, there is a critical demand for dynamic systems that can incorporate customer feedback with the entire levels of the supply chain. The existing static models lack the ability to extract consumer behaviour and market trends data in a fast-paced environment, requiring more responsive and agile systems. The multiple façades of current supply chain model, involving goods suppliers, warehouses, and retailer, showcased a significant lag in responding and integrating customer feedback, specifically regarding customer preferences, satisfaction, and product quality. The delays hinder direct long-term customer relationships, brand reputation, and operational efficiency. Hence, solving these problems will significantly be beneficial for both operational improvement and strategic necessity.

To address the previously mentioned issues, this paper proposed a solution that involves a sophisticated feedback system that leverages advance analytics and real-time data processing that can process customer reviews and incorporate knowledge to the entire supply chain network. The system leverages Large Language Model to extract the sentiment and suitable information for each supply chain participant out of the customer reviews effectively. The information taken will be distributed through the decentralised network supported by the blockchain to offer actionable alerts and insights to the entire supply chain participants. This integration will provide all entities an appropriate notification of the market changes and customer needs that allows them to be more strategic in inventory management and adjust their production line.

Therefore, the goal of the research is to develop a responsive supply chain that can quickly adapts to customer feedback to decrease the product returns volume and frequency, enhancing the products offering and customer expectations which eventually impacting the overall customer satisfaction.

The supply chain industry is afflicted with operational challenges due to the current unintegrated feedback loops in the supply chain and solutions have been proposed to fight such issues comprehensively. The impact of the solution implementation is expected to result in the significant reductions in operational inefficiencies, better products alignment to the customer expectations, decrease returns and dissatisfaction related cost, and eventually, a competitive and more robust position in the market.

1.4. Research Questions

An impactful study seeks to dive deep into the areas that can uncover insights and actionable solutions. In this research, the aim is to investigate AI's role to be integrated with blockchain in the supply chain, and how the technologies can provide insights from the customer feedback. Throughout the study, several research questions will be answered, such as:

RQ1. How LLM powered aspect-based sentiment analysis involvement in the supply chain increase the understanding and acting upon customer review?

RQ2. How can supply chain efficiency be improved by enhancing customer review sharing within supply chain components?

1.5. Aims

The aims of the research is to design and develop a revolutionary blockchain platform for supply chain management by utilising smart contracts with Proof of Authority protocol, inter-entity messaging protocol, and aspect-based sentiment analysis to enable real-time customer feedback analysis, therefore, will facilitate a transparent, efficient, and responsive supply chain network where transaction verification, data access control, and rapid customer feedback action can be done by the supply chain participants.

The significance of the study connected sophisticated technologies such as blockchain and LLM to the supply chain management field. The ability to share and analyse information in real-time, the research aims to decrease regular supply chain problems like inefficiencies, data silos, and inefficient feedback loops, that will result in product quality and customer satisfaction improvement.

1.6. Objectives

- To review the ability of AI and blockchain to categorise and identify precise customer review from various product and service quality aspect in the supply chain
- To determine the impact of AI and blockchain can immensely impacted the supply chain context sentiment that can help companies' decision-making process to focus on a certain area to be improve
- To build an integration platform that can ensure the accurate and safety of customer review for both AI sentiment analysis within a secure and transparent Blockchain framework.
- To examine the blockchain role in improving transparency and trust in customer review sharing to various companies within the supply chain network
- To quantify supply chain efficiency enhancement due to the improved feedback mechanisms thanks to AI and blockchain employment to manage and act upon customer reviews
- To assess the best practices for the technical execution and strategic integration of such technologies in supply chain management to improve profitability

1.7. Research Methodology (qualitative; 1 paragraph, quantitative: 1 paragraph, brief of 1st section of 3rd chapter)

In the Qualitative Analysis part, Talk about data collection, type of your data, and mention all details will be shown in Chapter 3

As a Qualitative Analysis, we use BC and LMM and this will be explained in Chapter 3

The research's applied methodology comprises the integration of multiple AI models and blockchain technology for sentiment analysis to push further the supply chain information sharing system improvement. The main focus is the efficient and responsive feedback mechanism development within supply chains with real-time customer secure data sharing and review analysis.

The research's main objective is to create and carry out a multiple AI models leveraged blockchain-integrated platform to enhance supply chain management. The objectives comprises of building a conceptual model to combine AI models and blockchain in supply chains, conducting model simulation development and validation to then be utilised for real-time customer feedback analysis impact on the supply chain efficiency, and modeling refinement based on validation results and scenario analysis.

The encapsulated domain, research goals, and major determinants of supply chain efficiency in the conceptual model, which include feedback mechanisms and customer reviews, blockchain technology for transparent transaction and secure data sharing, AI models to predict reviews' sentiment and category. Customer satisfaction, inventory control, product quality, and feedback loop efficacy are all the factors that impacted the supply chain efficiency. The representation of components' relationships is shown through flow diagram, which showcase the dependencies and interactions within the supply chain.

In the first step of the model development requires flow diagram creation that pictured the supply chain customer feedback, technological integration, and components relationship. The diagram includes key entities such as suppliers, warehouse, retailers, and customers. feedback loops and information flows are provided by blockchain and AI models, making sure that entire supply chain's aspects are correlated and responsive to customer feedback.

Next, components in the supply chain will be sorted as the second step, based on its ability to fulfill product demand, inventory availability, prices, and transaction records. The integrity of these records are ensured by the blockchain, keeping a transparent and secure system for data tracking and analysing.

Last, the third step involves simulation model development using Python to represent the conceptual model. This simulation model includes agent-based diagrams to illustrates the interactions of individual components (like suppliers, warehouses, retailers, and customers) within the supply chain. Agent-based modeling enables the presentation of each component's decision-making process and behaviour which enhance audience understanding of of supply chain's adaptive and dynamic nature.

Model validation was shown through multiple visualisations and the supply chain daily activities simulations for each payment, delivery, and product flow. The illustration presented by the visualisation shows each component (suppliers, warehouses, retailers, and customers)

interacts with each others to instill the model's supply chain dynamics representation accuracy.

The objective of the simulation is to capture the supply-chain management real-time processes, which include the time when each component delivers the products and makes or receives the payment. By using visualisations to track the activities, the model offer flow of goods and information clear depiction throughout the supply chain.

Simulations' result were evaluated and interpreted concerning the study goals. The key metrics of the evaluation is to visualise the enhancement through supply chain transparency and traceability, which will be very helpful to determine the blockchain and AI integration effectiveness approach to improve customer feedback responsiveness and supply chain management.

1.8. Tools and Techniques

Which tool or technique that you used? Put it in points

- Literature review to provide a holistic view on what people have been done in the area of (Technique)
- BC technology to model
- Semntiment Analysis to model
- Experimental Design
- Validation

1.9. Research Deliverables (the output of the "tools and techniques")

The research study will identify and try to investigate how blockchain technology can be integrated and used in studying the performance of information-sharing systems along the supply chains. In fact, this kind of research will be informative in showing and telling how blockchain could be merged into the management of a supply chain to optimize transparency, efficiency, as well as trust amongst stakeholders. These will contribute to the academic knowledge bases and also extend practical perspectives for industry professionals who seek to exploit blockchain in their supply chain operations.

The critical literature review of existing blockchain technology literature and its applications in supply chain management is the first key deliverable. It tries to give an indication of some current issues standing between traditional supply chain systems, such as data silos, lack of transparency, and inefficiency in general about information sharing. That is followed by discussion on how this may be addressed by blockchain technology through a decentralized, immutable ledger which will enhance data integrity and visibility along the supply chain. It will also indicate the gaps in the literature, hence setting a basis for the empirical investigations that follow.

The second key deliverable is the development of a conceptual framework showing how blockchain could be stitched into the existing supply chain systems. It will highlight the various elements constituting a supply chain, which typically consist of suppliers, manufacturers, distributors, and retailers, and show how blockchain technology has empowered this grouping to share information with minimal friction. Secondly, it will also discuss possible challenges in the adoption of blockchain resulting from scalability issues, regulatory challenges, and the necessity for interoperability with the existing system. This would be of immense assistance in giving organizations clear guidance on how implementation is to be done in the blockchain to help them in improving supply chain processes.

The empirical analysis will involve the case studies or pilot projects regarding the application of blockchain technology to supply chains. Case studies show tangible insight into benefits and challenges arising out of the application of blockchain in supply chains. Examples may include the study of how blockchain has been used in origin tracking, meeting regulatory requirements, or reducing fraud and counterfeiting risks. These findings will be useful in the case study for verification of the conceptual framework and providing fact-based recommendations to industry practitioners.

It will also, in parallel, elaborate on the best practices for effective blockchain implementation in supply chains. Such guidelines will be developed based on the literature review and empirical analysis to serve as actionable recommendations on common challenges such as data integration, stakeholder buy-in, and change management. Best practices shall be developed for different kinds of supply chains, considering the industry, scale, or geographical scope.

The research will finally be summarized in an extended conclusion where findings from the literature review, the conceptual framework, empirical analysis, and best practices are synthesized. This conclusion can thus provide an overall view of how blockchain is set to revolutionize supply chain management and outline the possible benefits, challenges, and future directions of research and practice. Deliverables in this research would, therefore, establish the laying of a solid foundation to further advance both theory and practice with blockchains in supply chain management.

1.10. Research Scope

The research scope investigates the application of blockchain technology in supply chain information-sharing systems to enhance the level of transparency, efficiency, and trust among diverse stakeholders who operate within supply chains. This present research has thus been limited to investigating the potentials of blockchain studies within supply chain management but most importantly for investigating data integrity and visibility and reducing inefficiency.

The research will essentially cover the current scenario of the supply chain information-sharing systems. The research study will delve into the existing challenges in the traditional supply chain models-fragmented data system, lack of real-time information sharing, and issues related to data security and authenticity. Understanding these challenges will position blockchain in a way that, with its decentralized and immutable ledger system, it ensures that accurate and current information is present to the participants in the supply chain.

In this regard, the research shall also address precisely those components of the supply chain where the integration of blockchain will most probably result in maximum impact. This involves origin-to-consumer tracking and verification of products, with assurance that all transactional data is recorded in a highly transparent and secure manner. Specific foci will be placed on industries such as food and pharmaceuticals, which have high needs for traceability whereby the authenticity and safety of their products could partly be certified through blockchain. By focusing on these key areas, the research will indeed provide very valuable insights into how blockchain can practically be put into place to enhance supply chain operations.

This research explores further possible benefits of blockchain, but also recognizes limitations and challenges accompanying its adoption in supply chains. The second set of barriers concerns issues of technology: scalability, for instance, and the huge computational power that blockchain networks demand; standardization between different blockchain platforms needs to take place. The second part deals with regulatory and legal implications due to the fact that in some cross-border supply chains, not all countries would have taken similar steps towards acceptance or regulation for its use. Through

these considerations, the research tries to bring a balanced view about the opportunities and hurdles towards adopting blockchain in supply chains.

In geographical terms, the scope is mostly global because supply chains often cross several countries and regions. However, specific case studies may be drawn from particular regions or industries where blockchain implementation is either in progress or in the state of ample potential for adoption. The study will, therefore, be supported by the incorporation of cross-comparison of these various case studies to determine good practices and lessons learned from different contexts that can then be applied to wider supply chain systems.

Finally, this work tries to give a conceptual framework showing how the blockchain technology would be integrated into supply chain systems. This, therefore, gives a guide to the entrepreneurs and policy makers on how this blockchain technology can be adopted, hence offering a roadmap in its implementation. The paper will also include practical recommendations for overcoming such challenges, ensuring the findings of the paper are theoretically valid and could be applied in a real-world scenario. A clear definition of the scope of research ensures that this study will be a valuable contribution to the field of supply chain management with more insightful details on the role of blockchain in industry transformation.

1.11. Thesis Organisation

1.11.1. Introduction

The chapter opens the way for the dissertation to introduce the key themes and technologies: Aspect-Based Sentiment Analysis and Blockchain. It showcases the main idea of the research problem and the importance of the technology integration to the supply chain. The chapter highlights the research questions, objectives, and potential implications. It also outlines the gaps in current methodologies and benefit of the technological application to improve supply chain processes and customer satisfaction.

1.11.2. Literature review

The literature review chapter explores and synthesises previous researches related to aspect-based sentiment analysis, blockchain technology, and the supply chain management implementation. It evaluates diverse methodologies employed to examine blockchain role to make maintain data transparency and integrity and customer feedback aspect sentiment categorisation. The chapter aims to provide a rationale for the research approach by connecting the theoretical concepts and previous empirical studies to research questions and objectives. It will pinpoint how findings in the previous research are related to the study and determine the gaps that the research wants to fill.

1.11.3. Methodology

The chapter defines and justifies the employed methodologies of the study. It highlights the technique used to apply AI and Blockchain in the supply chain industries, like data collection method and source, tools to build the technology, and the logic behind the leveraged methods. The chapter also describes the study design, involving experimental setups, the technologies deployment, and the evaluation metrics. Also, it also discusses the chosen methodology limitations, ethical considerations, and reliability and validity of the result.

1.11.5. Result

The Results chapter presents all of the results obtained from the experimentations and the application of new technologies into the supply chain setting. This reviews all performance metrics concerning AI models. Subsequently, the experimentation on blockchain technology has contributed towards ensuring data transparency and integrity within the supply chain. Critical insights are provided underpinning independent improvements in customer sentiment categorization accuracy and enhanced reliability of the supply chain operations through blockchain integration. The chapter also makes a comparison across different models and configurations to assess which kind of combination gained the most remarkable improvements in terms of alignment of the offered product with customer preferences, and a boost in overall supply chain efficiency.

Discussion

The Discussion chapter puts the results into a broader framework of the research objectives. Critical analyses of the implication that can be put forth or inferred from the findings are presented. The interrelations among the various outcomes of the study and their links with previous literature are drawn by explaining how AI and blockchain technologies integrated into supply chain management have improved customer satisfaction. This chapter also covers the limitations of the study, showing awareness of its potential effects on generalizability and suggesting topics for further investigation. The discussion gives in-depth understanding regarding the importance of the research and what it contributes to the discipline through this analysis.

Conclusion

The Conclusion chapter summarise the key findings and contributions of the research, emphasizing once more how AI—blockchain technologies integration has improved supply chain management and enhanced customer satisfaction. Furthermore, it revisits what was introduced as the main objectives of the research in the introduction and affirms that they are met based on the study's outcome. In addition, the chapter shares the practical implications of the research and, at the same time, does not fail to indicate its limitations. The conclusion gives the last comments on the impact of the study and points toward the future trajectory of research, ensuring a full closure to the dissertation.

2. Literature Review

To be able to excel in the competitive market, companies must be able to perceive customer demands and convincing the market to purchase their product, however, most of the data provided by the market are intangible, meaning, the data cannot be easily measured like numerical data. For example, online customer reviews are invaluable for various organisations' sectors to provide actionable insights that push business decisions strategically. By enabling reviews analysis helped by Aspect-Based Sentiment Analysis models, companies will be able adjust their corporate strategy to anticipate future demand.

- a. Previous work on sentiment analysis and LLM within supply chain customer review analysis

Amellal et al (2024) conducted advanced analytical models integration, like Bidirectional Encoder Representations from Transformers (BERT), used for analysing text data, Gated Recurrent Unit (GRU), for forecasting demand, and Bayesian Networks, offers pricing strategies probabilistic insights, has been proven to significantly improved the abilities of the supply chain management in the unified platform through the real-time and comprehensive overview enablement of multiple important components. Both supply chain operations' micro and macro aspects will gain the clarity of an unprecedented decision-making level thanks to this system integration.

More valuable consumer sentiments, pricing dynamics, and demand forecast knowledge can be gain by businesses through the holistic platform utilisation. By applying BERT to the customer sentiment analysis offers a subtle consumer preferences and emotions comprehension. The knowledge enables firms to suit the product offering and marketing strategies better suit the wants and demands of the target audience that will increase the customer loyalty and satisfaction. Also, the instantaneous analysis of BERT ability leads corporates to adapt their positioning to the changing market conditions and consumer trends more quickly.

Furthermore, the integrated system was also been tested on real-life like simulations that mirrors the modern supply chain dynamics and complexities. Realistic scenario mirroring that allows system's accuracy and responsiveness assessment are made possible due to the diverse type of data input that include market trends, historical data, logistical data, and social media feedback, that enables it to mirror a realistic scenario to assess system's accuracy and responsiveness, thanks to its capability to handle large-scale data, predicts outcomes, and adaptability to changes. For example, BERT's sentiment analysis ability enables the system to detect a shift in consumer sentiment of a product demand, which will be responded by the GRU model by updating its forecasting approach due to the changes, followed by the Bayesian Network to amends the pricing strategies in response to the anticipated demand changes.

Furthermore, Capuano et al (2020) made an effort to apply sentiment analysis to the Customer Relationship Management (CRM) systems was explored by leveraging sophisticated machine learning models by employing Hierarchical Attention Networks (HANs) that utilise incremental learning capabilities to push further the CRM systems interaction and feedback processes. The integration to the CRM system let the model to adapts and evolves with the help of the real-time feedback, allowing a dynamic approach to improve its accuracy and effectiveness continuously in customer sentiment analyst.

The model utilises an attention mechanism that is capable at focusing and identifying most valuable information of a lengthy text, which does not necessarily require the entirety of the text to be used by the model to conduct the sentiment analysis. For example, when the customer review discusses multiple product aspects, the attention mechanism distinguishes and rank the expressed critical features sentiment from less relevant content to the most relevant, which is suitable for the purpose of CRM system supported by the selective focus to precisely analyse the correct sentiment, by forcing the attention layers to work in conjunction

to score and weigh different parts of the input data. Each layer works of diverse level of data, from each individual words to the whole sentence or the entirety of the paragraphs, strengthening its ability in handling complex and varied data structures, which impacted its sentiment detection accuracy and effective customer feedback nuances and context capturing.

Also, the model incorporated incremental learning that addresses the new information challenge without eliminating the previously gained valuable insights from historical data. The dynamic data taken from CRM systems can be well-managed by the incremental learning approach, due to the everchanging market conditions and customer preferences. New continuous updates in parameters can be conducted with the help of incremental learning capability when taking new data, maintaining high accuracy level and relevance over time.

On the other hand, García-Pablos et al. (2017) used an unsupervised approach to retrieve the sentiments out of a review, which eliminates the need to label each dataset that is extensively time consuming like when supervised learning model was employed. Word embedding with Latent Dirichlet Allocation (W2LVA) is the unsupervised model that was leveraged to improve the ability to categorise and discern text data aspects. It surpasses the limitation of traditional supervised model by using the innovative system that decrease the requirement to manually labelling the data, by allowing the model to understand the words semantic relationships that showcases the more nuanced insight and language processing. The Word Embedding aids with the word mapping to a high-dimensional space based on their similarities contextually, and the LDA, a well-known topic modelling method, also helps with identifying hidden topics in the texts that match the products or service aspects being discussed. The incorporation of the two method enables W2LVA to enhance the parsing and categorising of sentiments without the need of extensive human oversight in user-recorded content.

Diverse datasets from various industries and languages, involving English, Spanish, French, and Dutch, was utilised to rigorously test the ability of W2VLDA, and benchmarked with multiple sentiment analysis methods that also have the capability to effectively classify and discern the aspect related sentiment throughout the texts. Ultimately, W2VLDA managed to exhibit its superior adaptability to the new languages and domains with enhanced accuracy shorter time and cost, which is required if traditional sentiment analysis were employed, and legitimate its potential to be considered in global market analysis as a versatile tool, making it a more efficient and scalable solution to keep the pace with the quickly growing online content.

Bagheri et al. (2013) utilised a generalised method that allows multi-word aspects learning was also introduced to address the challenges that cannot be done by supervised models. A set of heuristic rules that take into consideration the opinion words influence, the words that define the sentiments like “good”, “poor”, or “love”, on aspect detection was added to the method, allowing the text relevant aspect identification accuracy improvement. Moreover, a mutual information and aspect frequency based novel metric was leverage that helps with the aspects scoring. The metric eliminates the challenges faced by supervised models, such as labelling the training data, due to its adaptability that function with a set of unsupervised

seed within a new bootstrapping iterative algorithm. Additionally, two pruning methods was incorporated with the purpose to get rid of wrong aspects in relevance to the correlation of the identified aspects in reviews, which is one important refinement to maintain the aspect detection process accuracy and relevance.

As the result, the model can extract the implicit aspects based on the association of the explicit aspects with its opinion words. For example, a product review that stated “An excellent phone that can last the whole day” has the “phone” as the explicit aspects, while “excellent” is the opinion word, the model had frequently paired “excellent” with “battery life” by the co-occurrence metric to comprehend that the positive sentiment is probably related to the phone’s battery, although “battery life” was not stated. During the rare cases where the model incorporated the sentiment with the wrong aspects, firstly it prunes the aspects that are rarely mentioned or do not fit well with the others in relation to the core product aspects, like if “screen brightness” is barely mentioned feature within smart phone reviews, and then, prune the aspects even further if it may confuse or contradict the contextual relevance to avoid inaccuracies, thus completing the two step pruning methods. By mapping the pre-extracted explicit aspects to each opinion word in the extracted polarity lexicon, the model can achieve this ability.

Lau et al. (2018) utilised a parallel aspect-oriented sentiment analysis integrated method of novel big data analytics that improve supply chain management sales forecasting by using an online platforms consumer interactions recorded extensive data that targets the product reviews and comments expressed sentiments. By employing the parallel aspect-oriented sentiment analysis algorithm to the model, capable of processing huge number of datasets effectively that focuses on consumer reviews specific product aspects. The various product aspects like feature, quality, or price, will then be detected and categorised by the sentiment analysis component through the related sentiments, which enables a more profound comprehension to the customer reviews and the sales predictions impact. By addressing the traditional sales forecasting methods limitations as the primary goal that often unable to understand nuanced consumer sentiments expressed in diverse digital platforms. It is expected to increase sales forecast accuracy through the sentiment analysis implementation, which can be helpful for businesses to effectively manage their inventory and better respond to the trends and needs of the market.

Through the use of big data processing tools and advanced machine learning algorithms for large data volumes analysis and handling. The method employing parallel processing techniques to help control the extensive datasets sentiment analysis computational demands. With the help of the aspect-oriented analysis, the model can focus on products distinct aspects related specific sentiments that were extracted and identified from the online reviews text. The research was tested on a real-world scenario by using product sales dataset to empirically test the model effectiveness. The output shows noteworthy enhancement in sales forecast accuracy with the addition of consumer sentiment data. The model ability to better align product availability with consumer demand in actual scenario implementation showcases its efficiency and practicality in customer satisfaction enhancement and inventory mismanagement risks reduction. Despite the model returns a sophisticated model that

integrates sales forecasting and sentiment analysis, a deeper study on the associated obstacles with across various platforms diverse data quality. The consumer expression variations and online reviews inherent biases can impact the sentiment analysis accuracy. In addition, the research also lacks a sufficient data treatment method against noise that is common in large-scale social media feeds.

Durairaj & Chinnalagu (2021) experimented with an enhanced mixed-language texts sentiment analysis hybrid deep learning model was developed, which will be beneficial for the customer reviews where multilingualism is commonly occurred, significantly enhancing the ability to analyse the ever-growing varieties of languages in customer feedback volume that will cannot be done with conventional sentiment analysis models. The hybrid model incorporates Sequential Attention-Based Long Short-Term Memory (SAB-LSTM) network to the fastText word embeddings. Firstly, fastText efficiency was utilised in semantic meaning capturing across different languages and sub word information handling, showcasing its highly impeccability in processing morphologically rich languages. Secondly, the SAB-LSTM leverage an attention mechanism for text part prioritisation based on the relevance of sentiment determination, which helps improving the sentiment prediction accuracy of the model. The researcher's main goal for the development was to address the obstacle presented by the mixed-language sentiment analysis texts. Highly accurate sentiment interpretation of non-uniform linguistic data often met by failure by tradition models due to its contextual variations and linguistic nuances effective handling incapability. The goal was to develop a more reliable model that eliminates the extensive pre-processing or language-specific tuning and still capable to produce an accurate multi languages sentiment analysis.

The research methodology leveraged the fastText application as word embeddings generator for effective linguistic features capturing in the texts, integrated to the enhance LSTM network equipped with attention mechanism to focus on significant parts of the texts. The integration enabling the dynamic adjustment of the different text segments focus based on the expressed sentiment relevance. The model evaluation utilised mixed-language customer reviews dataset from multiple online platforms by comparing several sentiment analysis models including SVMs and LSTM networks. The output presented that the hybrid model outperformed the traditional model in mixed-languages complexities handling effectiveness and sentiment analysis accuracy. However, the research could be improved by the used of wider types of mixed-language combination examination and deeper exploration to the impact of configuration changes in the LSTM network's attention mechanism.

Alqaryouti et al. (2020) built a hybrid aspect-based sentiment analysis model that connected domain-specific rules and lexicons. The model is specifically meant to be used for smart app reviews analysis offered by government entities. The main development emphasises on improving the public sentiment interpretation and categorisation capacity to be more accurate for the services provided by the government. The model utilised both rule-based approach and machine learning techniques, integrating domain-specific lexicons that are specifically tailored to government services. The incorporation of enable text data nuanced comprehension, allowing the classification and identification of the linked sentiments with various service aspects mentioned in user reviews. The main goal of the

model development was to solve the parsing and interpreting obstacles of multifaceted and complex government app reviews public sentiment, which can be difficult to be done with traditional sentiment analysis tools to understand the context-dependent and specificity nature of language in such reviews. The model is expected to offers an analysis with better precision by integrating domain-specific terminology and context understanding, hence, improving governmental and transparency of public feedback response.

To accomplish the research aim, a combination of Natural Language Processing (NLP) tools and innovative application of rule-based parsing and domain-specific lexicons were involved. The methodology makes sure the automation of sentiment analysis but also specifically suited to the government services unique context. The lexicons and rules were built based on comprehensive examination of expressions found in the review data and common terminologies, and then encoded to the system which increase the relevancy and accuracy. Then, the model was implemented to a government app reviews dataset that included diverse feedback on various aspects of the functionality and service quality of the app. The result showcased inherent enhancement in the detection accuracy and sentiment classification ability. The improved model offered deeper knowledge into areas of concern and satisfaction of the users, enabling targeted responses from the government entity. Yet, the paper did not take into consideration about the scalability and limitations of the model. The research can be benefited by wider exploration of the model performance on various of datasets. Model adaptability to handle the perpetually changing linguistic terms and trends was also not covered in the research.

Uma et al. (2022) came up with a primary development that performs customer review sentiment analysis using the SVM and CNN models application that are designed to conduct sentiment classification based on the text data into positive, negative, or neutral categories. High-dimensional space handling and optimal hyperplane finding proficiency of the SVM model was used to classify data points, while the CNN model's ability to identify sequence data patterns that is suitable for text data was leveraged. Textual content semantic information and local dependencies of the text data can be captured by these models which will enhance the sentiment classification accuracy and efficiency, the main goal of the research. Currently, the methods of sentiment analysis faces challenges in interpreting human language complexities, involving context dependency, sarcasm, and implicit meaning. The research aims to polish the processes for sentiment classification to offer a more trustworthy analytical tools for businesses to understand consumer opinions accurately.

Several methodologies like vectorisation, tokenisation, and normalisation were leveraged to preprocess the text data to be fed to the machine learning models. The research split train and test data to review the models impartially. Furthermore, performance metrics including accuracy, precision, and recall are computed to evaluate SVM and CNN models effectiveness. As a result, both models return high accuracy rates, where SVM's accuracy is slightly below CNN model. The superior performance of CNN in text data handling suggest that its local feature dependencies processing in sequences capability provides a significant enhancement in sentiment analysis. Although the paper offers a diverse analysis of the CNN and SVM models in sentiment classification, it does not provide a detailed discussion on the model

limitations, especially in nuanced linguistic features handling like jokes or irony that are common in customer reviews. Also, the imbalanced datasets impact on the model performance was not explored in the study. Furthermore, the study could be benefited by the other state-of-the-art model comparisons like BERT or LSTM to offers better image of the relative performance of the models and better developed CNN case and ability in sentiment analysis.

Ng et al. (2023) designed an empirical evaluation framework for diverse machine learning and deep learning models comparison for e-commerce platform customer reviews sentiment analysis. The goal was to determine the most effective models to interpret user-recorded textual data complexity. The research particularly focused on evaluating multiple advanced models, like RoBERTa, LightGBM, and BERT. Each model was assessed based on the text sentiment categories classification accuracy capability. RoBERTa and BERT are models that are based on the architecture of the transformer, notable for its ability in deep contextual learning and are important for customer reviews nuance comprehension. The comparative study was driven by the necessities to automate e-commerce customer reviews sentiment analysis that can offer impactful insight through customer opinions and allows business to improve their decision-making processes. The goal was to detect the most trustworthy and accurate model for the sentiment analysis purpose to be tailored to the business strategies to better understand customer preferences and needs.

The author employed methodology include customer reviews dataset collection that will be processed and analysed by the models, also, performance metrics like accuracy, precision, recall, and F1-score to assess model's effectiveness to be then compared based on their strengths and weaknesses in the real-world implementations. By leveraging large customer reviews taken from popular commerce platform that consist of customer opinions and feelings with diverse expressions ranges from disappointment to satisfaction. Consequently, the traditional models like LightGBM and the others were outperformed by the transformer-based models, RoBERTa and BERT, based on all the key performance metrics. The superiority was made possible thank to the capability to process and analyse text data with deep context and semantics comprehension that is important to classify customer reviews sentiment accurately. However, the research could be improved by a deeper exploration of different e-commerce platform types and domains and its effect on the model's performance, despite having a sufficient insight into the sentiment analysis models effectiveness. Also, increasing the varieties of the dataset that contain reviews from diverse sources globally to evaluate the robustness and adaptability of the model to varying cultural nuances and languages. Lastly, the resource utilisation and computational efficiency inclusion in the assessment can offer a better sufficient understanding of the models' deployment practical implications in real-world business environment.

Pecan et al. (2018) investigated the various text pre-processing techniques impact on the neural network model performance at analysing customer reviews. The contribution of the study to the body of knowledge by clarifying the significance of pre-processing steps in improving the sentiment analyses accuracy. The focus of the study development circles around neural network model that is tested with variations of configurations of text pre-

processing techniques. These techniques involve stop words removal, stemming, and tokenisation. The model is designed to evaluate how these text modifications impacted the ability of the neural network to classify the expressed sentiments in customer reviews accurately. The main goal of the development is to showcase the underexplored area in sentiment analysis text pre-processing, which demonstrates further that a well-prepared and cleaned text data can substantially affect the sentiment analysis result, stemming from the hypothesis that more efficient pre-processing techniques can improve data quality and reduce noise, hence, improving the performance of the model.

During the methodology employment, a series of pre-processing techniques combination experimentation are implemented to the customer reviews dataset. The performance of neural network model with each combination is assessed to detect the effect of the techniques, which then showcase the effectiveness of isolation and combination of each pre-processing step comparative analysis. The output of the study determines the pre-processing techniques, specifically stop-word removal and stemming notably enhance the ability of neural network to accurately classify the sentiments. The research shows that these methodologies allow the model's attention to focus on relevant features by eliminating irrelevant and redundant data, which increase the overall sentiment classification accuracy. Despite the provided beneficial insights of the text pre-processing sentiment analysis impact, the various neural network architectures and various pre-processing techniques interaction was barely discussed. In addition, the research can be benefited by a more rigorous analysis of the impact of such techniques to the varieties of customer reviews languages and culturally nuanced expression that can vary widely and affect the results generalisation, which limit its applicability.

AUTHOR, YEAR	FACTORS	AIM	METHODOLOGY TYPE
(Amellal et al., 2024)	Supply chain management advanced models	Leverage sophisticated models such as GRU, Bayesian Network, and BERT across supply chain management aspects	Conceptual/Theoretical
(Garcia – Pablos et al, 2017)	Aspect-Based Sentiment Analysis	Built an almost aspect-based sentiment analysis unsupervised system for various domains and languages	Unsupervised learning
(Capuano et al, 2020)	Customer relationship manager	Enhance CRM using Hierarchical Attention Networks-based model that integrates incremental learning	Model Development/implementation
(Bagheri et al., 2013)	Sentiment analysis	Suggest an unsupervised, domain-independent model for sentiment analysis aspect detection	Unsupervised Learning/ Model Development

(Lau et al., 2018)	Big data sales forecasting	Improve supply chain management sales forecasting by parallel aspect-oriented sentiment analysis	Big Data Analytics
(Durairaj & Chinnalagu, 2021)	Mixed-language customer reviews	Analyse mixed-language customer reviews sentiment through a novel hybrid deep learning model	Hybrid Deep Learning Model
(Alqaryouti et al., 2020)	Government review data	Implement a hybrid aspect-based sentiment analysis model to government own smart app reviews	Model development/implementation
(Uma et al., 2022)	Customer review sentiment analysis	Customer review sentiments classification leveraging SVM and CNN models	Machine Learning/Model Development
(Ng et al., 2023)	E-commerce platform customer reviews	Machine learning and deep learning models comparison involving LightGBM, RoBERT, and BERT for sentiment analysis	Comparative Study/Model Evaluation
(Pecan et al., 2018)	Text pre-processing impact on sentiment analysis	Text pre-processing techniques effect on the sentiment analysis performance study	Experimental/Model Evaluation

b. Previous work on blockchain applications in supply chain management

Azzi et al. (2019) constructed a conceptual framework to incorporate supply chain management (SCM) and blockchain technology to increase the reliability, security, and transparency of the supply chains that are commonly faced by problems like tampering, corruption, and fraud in centralised systems. The model utilised distributed ledger technology in the blockchain to verify that each supply chain transaction is immutably recorded to enable cost reduction and efficiency reduction related with SCM. The objective of the research is to find the solution to the traditional supply chain systems obstacles that includes high susceptibility to fraud, centralisation inefficiencies and lack of transparency. Through the integration of blockchain, the research aims to develop a more efficient and reliable system that can record the transactions transparently and share information securely.

By combining both theoretical examination and real-world application analysis, the authors explain the basic characteristics of blockchain, like immutability, transparency, and decentralise, and describe their implementation in SCM. The conceptualisation from the framework uses smart contracts to automate the processes and reduce dependence on intermediaries and manual checks. Although the research does not provide a detail measurable empirical case study, it offers a theoretical analysis of revolutionary potential

brought by blockchain in SCM. It anticipated enhancement and fields such as counterfeit and grey market trading loss reduction, goods traceability, and improving supply chain partners reliability. The paper offers significant insights in the supply chain reshaping potential by blockchain. But the empirical evidence that support the theoretical claims is absence. The research can be benefited by including pilot projects or case studies that presented the application of the proposed blockchain-based supply chain model. Moreover, the research disregards a sufficient solution for the blockchain high energy consumption and scalability obstacles.

Du et al. (2020) developed a platform that utilises blockchain system to assist financial seamless and transparent transactions within the supply chain participants to better enhance the general trustworthiness and efficiency of the network. The components of development in the study in the study involved the blockchain technology integration to the financial platform used in the supply chain. The platform ensures transparency and security of the transaction processes in the entire supply chain from the beginning to the end. Also, smart contracts that automate and record transaction immutably and securely was introduced to the blockchain architecture. The main goal of the development is to build blockchain-based supply chain finance solution that can solve issues like inefficiencies, trust, and traditional supply chain finance high cost. Issue like centralisation in the traditional models often leads to the potential of fraud and lack of transparency. However, blockchain's nature of decentralisation can solve such problem by offering a tamper-proof system that records transactions that reduce the possibility of fraud and improve trust.

The employed methodology used in the study include tailoring blockchain architecture with supply chain finance, which involved the utilisation of homomorphic encryption that make sure the privacy of data in the blockchain. The implementation of smart contracts automates the financial obligations and transactions within parties. Rather than a practical application case study, the study offers a theoretical evaluation that discusses the possible enhancement that can be provided by the proposed blockchain solution, like increased transparency, advanced security, and more efficient capital flows, which can eventually lead to costs reduction and improve financial services for supply chain participants. Despite of all the outlined blockchain benefits in the research, it lacks the empirical data that can support the claims. The research only covers the theoretical side and disregard the implementation or cases in the real-world that can showcase the practical effectiveness and blockchain solution deployment challenges. Also, although the research discusses data privacy through homomorphic encryption, it barely explored the scalability issues potential of a blockchain systems, especially in large-scale and complex supply chain environments. Further study can consider practical challenges exploration and offers a more detail comprehension of the impact of blockchain in supply chain finance.

Koirala et al. (2019) Built a blockchain-enabled model that manages supply chain reverse auction bidding process. The main innovation showcased is the blockchain technology incorporation to improve the efficiency and transparency of the supply chain procuring carriers reverse auctions. The model integrates a deployed smart contract in the blockchain system that can record and automate bidding process transparently. The model was used to

conduct multi-attribute decision-making handling that allows buyers to assess carriers based on diverse criteria beyond price, like delivery time, reliability, and service quality. The goal of the development is to dismiss the barrier correlated in the traditional supply chain reverse auction processes, like collusion risks, complex bidding criteria handling inefficiency, and transparency. Through the utilisation of blockchain, the model aims to make sure that the entire decisions and transactions are immutable and transparently recorded, decreasing the fraud potential, and increasing trust within the participants.

To build the decentralised blockchain platform that enables supply chains reverse auction automation, the smart contract is in charge on the auction execution automation that allows all bids to be securely recorded and auction rules are obeyed without involving central authority. The research leverages simulation to test the proposed model in the hypothetical scenario based on a published case study data on the dairy supply chain reverse auctions. The simulation output presented that the blockchain-enabled model can improve the reverse auction process transparency and efficiency significantly. The smart contract provided a competitive and fair bidding environment, and the immutable transactions record enhance participants trust. The blockchain model efficiently take case of the multi-attribute bidding process that enables a more holistic evaluation of bids beyond price. Although the research extensively explored the blockchain application contribution to the enhancement of reverse auction processes, it lacks the consideration for its solution scalability and applicability in the real-world scenarios other than the simulated one. In addition, the research can be benefited by a deeper understanding of the integration challenges potential and overhead costs related to the blockchain implementation across complex and diverse supply chain networks. Furthermore, the environmental impact of blockchain application, such as the high energy consumption was not explained.

Saberi et al. (2018) research an extensive examination of the blockchain application to be used in the sustainable supply chain management (SCM). The paper studied the blockchain potential to improve traceability, transparency, and security in the supply chains, especially in the sustainability realm. The study focuses on the critical assessment of blockchain technology's features that are relevant to SCM rather than building a new one. It examined the blockchain characteristics, like consensus-based transaction validation, immutable ledger, and decentralised management that can inherently replace traditional supply chain operations. The main objective of the research is to deep dive how blockchain solve sustainable supply chain management issues like traceability inefficiency, security vulnerabilities, and lack of transparency. The research aims to show the transformative solution that can be offered by the blockchain to meet the ever-increasing sustainability demands in supply chains.

Qualitative approach was leveraged by collecting huge number of literatures to analyse the impact of blockchain to SCM. The research used systematic industrial resources and academic review to examines practical and theoretical aspects of blockchain application to offers detailed exploration of the challenges and benefits of blockchain potential. The study discusses blockchain's potential impacts and theoretical applications in SCM despite not focusing on any specific case study with empirical data. It confirms that transparency and

traceability can be significantly improved by the technology that is important to verify the supply chain sustainability practices and complying the social and environmental standards. However, the paper lacks any specific case studies or empirical evidence that showcased real-world implementations and blockchain outcomes in SCM. The limitation caused by the absence of empirical data caused the research inability to fully present the finding practical applicability to translates theoretical potentials to practical benefits. Also, the research can provide more detail discussion on the blockchain environmental impact and technology scalability, due to its nature to intensively consume energy and how it aligns itself to the sustainability goals.

In the agriculture and food (Agri-Food) supply chain realm, Shahid et al. (2020) contributed to the development of a blockchain-based solution sophistication by combining the smart contract and blockchain to ensure trust, security, and traceability of the supply chain network. The aim of the development addresses the trading processes accountability, products traceability, and entities credibility. Several components were included to work together in the proposed system to optimised Agri-Food supply chain. First, Ethereum blockchain network was utilised to deploy smart contracts to manage interactions and transactions in the supply chain. Then, blockchain recorded every transaction and the correlated data and make sure its transparency and immutability. Second, it integrated of Interplanetary File System (IPFS) to the system for secure and efficient data storage. IPFS returned a unique hash code for stored data, which then stored in the blockchain, connecting transaction record with the product information detail. The paper primary goal is to eliminate the traditional centralised supply chain systems limitations, which often plagued with lack of traceability, third-party intermediaries' dependence, and transparency issues. By applying blockchain based system, it can solve all these problems by showcasing the entire journey of Agri-Food products from the origin to the consumer, transaction transparency, and trust among stakeholders while eliminating the intermediaries.

Ethereum blockchain act as the deployment backbone by offering smart contracts deployment platform to secure and automate transactions. The IPFS integration make sure large data sets are efficiently stored while maintaining the accessibility and integrity of data. The paper provides a case study that include blockchain-based Agri-Food supply chain system simulation when being applied to hypothetical scenario. The output presented significant enhancement in several areas, such as product traceability increase, overall supply chain efficiency improvement, and fraud and errors reduction. The system also demonstrated potential costs reduction related to data management and logistics, hence, increasing the stakeholders' profitability. Nonetheless, the paper also disregarded the scalability challenges of such large-scale deployment. And, potential regulatory and legal challenges of food safety and data privacy in different jurisdiction are absent.

Gonczol et al. (2020) conducted surveys to understand the blockchain implementation and adoption in multiple supply chain sectors. The research conducts an extensive academic and industrial implementation reviews to present blockchain's integration and impact within supply chain. Several categories were extracted from the blockchain application in supply chain survey finding, such as potential benefits, range of applicability across different

industries, and technological features. The research offers a methodical break down of the blockchain application to categories that presented the maturity and industry adoption levels of the technology. Lack of a consolidated reviews that surpass both practical and academic blockchain implementations in supply chains is the primary goal of the research. By collecting the data through the survey, the research is expected to highlight blockchain transformative potential in advancing security, efficiency and transparency through current use, developments, and trends of the technology in the field.

By employing systemic review of existing literature and the present-day industrial practices analysis, the research collected data from conference proceedings, real-world blockchain implementations case studies, and peer-reviewed journals and analyse it with qualitative approach to asses the challenges and benefits of the integration of blockchain within supply chain industry. The paper does not focus on specific case study, but it combined diverse literature case studies involving industries like food, pharmaceutical, and manufacturing to assess the technology ability to ensure food traceability and safety, and combating drugs and manufacturing counterfeit. The research found that blockchain can reduce counterfeit and fraud risk, and eliminate intermediaries' reliance, which help to streamline the operations, due to its significant impact on enhancing transparency, efficiency, and traceability of the supply chains. However, the research be benefited through the discussion of the technology scalability problem and limitations. Furthermore, quantitative analysist would be preferable compared to qualitative approach to improve the credibility of the research. Also, a detailed discussion of the Blockchain and current supply chain IT system integration could help the organisations further on the blockchain adoption.

Centobelli et al. 2021 proposed "Triple Retry" framework approach that incorporate circular supply chain and blockchain technology to heighten trust, transparency, and traceability. The framework was created to cater the circular supply chain management complexity that includes the reverse logistics acquired for remanufacturing, redistribution, and recycling processes. The Triple Try framework presented some key components to utilise blockchain technology capabilities such as blockchain integration to provide an immutable ledger for assets tracking and transactions recording, smart contracts to automate and enforce contract terms and conditions in varieties of reverse supply chain stages, traceability mechanisms that ensure quality and environmental standards compliance by enabling detailed tracking of products and materials, and transparency tools that allows every stakeholders to view the supply chain processes transparently. The main purpose of Triple Try framework development is to solve stakeholders trust issues, product and materials lifecycle tracking difficulties, and transparency in the circular supply chains, which often inhibit the recycling and remanufacturing processes efficiency and effectiveness.

The research methodology consists of framework designing based on the previous theoretical and literature underpinnings to conceptualise the blockchain integration to circular supply chains while also implement smart contracts to maintain data integrity and process automation and assess its impact on the supply chain traceability, transparency, and trust through qualitative approach. The study does not include any empirical case study and only focus on the development of the conceptualise framework. However, the framework

implementation is expected to bring several circular supply chains enhancement, like increase accountability and traceability of products, improved stakeholders trust, and enhanced regulatory compliance. Although the Triple R framework offered blockchain integration novel approach in circular supply chains, the lack of empirical studies limits its practical effectiveness demonstration. The inclusion of pilot project that applies the framework in real-life scenarios could immensely improve the research by providing insights like potential operational benefits and challenges. Also, the research could explore the framework scalability and adaptability to various industries.

Kouhizadeh & Sarkis (2018) discussed the blockchain integration to greening supply chains, which help to conceptualise its impact on the environmental sustainability in supply chain processes, focusing on the synergistic potential to solve environmental issues. Some key components were addressed in the paper where the green supply chain management intersects with blockchain technology, such as transparency to showcase barrier-less view of the supply chain activities to the entire stakeholders to maintain the green initiatives credibility, decentralisation to decrease the central authority reliance that can potentially mismanage and obscure environmental data. The premier purpose of the study is to identify the critical gaps in the existing supply chain practices that affect environmental study due to the lack of environmental claims authenticity, sustainable practices tracking and verifying inefficiencies, and transparency.

The methodology employed theoretical and conceptual approach, collecting studies from many different supply chain management fields' literature, environmental sustainability, and blockchain to synthesize the knowledges of blockchain and green supply chain practices integration. Yet, the paper talks about the hypothetical scenarios and blockchain applications potential in greening supply chains while not providing any case study with empirical data. As the result, significant improvement due to the transparency and traceability enhancement in sustainability performance can be done with blockchain that leads to better environmental regulations and standards compliance. Despite the valuable theoretical foundation of the blockchain applications in greening supply chains potential understanding, it does not offer any empirical support or evidence to the claims. Most of the discussion are speculative and can be enhanced by actual pilot projects or case studies to demonstrate the implementations and advantages of the technology in real-world sustainable supply chain management. And, subjects like blockchain technical limitations and challenges including environmental impact and scalability can be further explored by the paper, especially regarding the large energy consumed by the technology.

A conceptual framework that delves deep into the Reverse Supply Chain Networks (RSCN) and Blockchain that improved multiple industries sustainability performance was developed by Jraisat et al. (2022). The research is helpful to bridge the gap in the technology implementation in circular economic models, accentuating blockchain role as a strategic tool to enhance reverse logistics sustainability practices. The frameworks concentrate on the dual incorporation of blockchain and reverse supply chain processes that includes blockchain application in accountability, transparency, and traceability throughout the process. Key components involve recycled materials verification, remanufactured products

authentication, and returned goods tracking. The main objectives of the development are to solve reverse supply chain issues such as resource utilisation inefficiency, sustainability authentication difficulties, and lack of transparency. Through the blockchain integration, the researcher aims to streamline processes, enhance reverse logistics environmental and economic sustainability, and foster stakeholders' trust.

Qualitative research design was employed on the study by leveraging both primary and secondary data from four distinct industry case studies such as toys, pharmaceuticals, food, and electronics. The primary data were collected through interviews with RSCNs experienced managers and technology applications, helped by the secondary data to determine blockchain and RSCN integration drivers. The involved four different industries showcased various blockchain applicability in reverse supply chain networks for sustainability enhancement by focusing on the blockchain practical implementation to manage and trace the product return, recycling, and remanufacturing processes. Key results pinpointed that blockchain significantly enabling the items reverse logistics journey followed through accuracy that enhance the sustainable practices accountability and stakeholder trust. The research does not explicitly mention the blockchain scalability within reverse supply chains despite the insightful blockchain benefits and strategic importance in improving reverse supply chains sustainability performance. The potential issues of blockchain technology scaling in various regulatory environments and technological infrastructures are important for broader applicability evaluation and limitation in reverse supply chain technology.

Chang & Chen, (2020) offered a systemic blockchain and supply chain management literature reviews that aims to connect the current industrial and academic research to breakdown of the blockchain withing supply chain field trajectory. The paper categorises current research into several focuses like technology, trade, traceability/transparency, and trust in the blockchain-implemented supply chains. Each category showcases specific part of blockchain implementations, like trade operations improvement, supply chain stakeholders trust facilitation, and transparency enhancement. The main objectives of the literature review are to combines the separated nature of the blockchain application in supply chains research. It aims to determine the current implementation, issues, benefits, and challenges in the blockchain-supply-chain paradigm, offering a comprehensive guideline for future research and practical implementations.

Comprehensive literature survey was conducted followed by analytical review that utilised methodological approach to categorise and analyse co-citation analysis and inductive reasoning to filter trends and key themes from the literature. Hence, no case studies were employed other than the one taken from the existing reviewed literature findings. The results show blockchain important role at solving compliance, efficient resource management, and counterfeiting problems by its transparency and traceability that can greatly impact the trust in supply chain. The review provides a thorough blockchain application in supply chain landscape mapping, but it can be improved by exploring blockchain adoption barriers in different regions or industries. Also, the research can discuss the environmental and scalability issues of the technology. Furthermore, blockchain types comparison analysis (e.g. public vs private) in the specific supply chain context would add value, offering stakeholders

better decisions making process to decide which blockchain configuration serve their needs the best.

AUTHOR, YEAR	FACTORS	AIM	METHODOLOGY TYPE
(Du et al., 2020)	Supply chain finance	Solve non-trust issues, information and capital flow enhancement, cost reduction	Conceptual/ Implementation
(Koirala et al., 2019)	Supply chains reverse auctions	Reverse auction bidding processes transparency and efficiency improvement	Simulation
(Saber et al., 2018)	Sustainable supply chain management	Study blockchain potential in traceability, transparency, and security improvement	Literature review
(Azzi et al., 2019)	Supply chains – Blockchain integration	Develop a transparent, reliable, and secure supply chain system	Conceptual/ Theoretical
(Shahid et al., 2020)	Agri-food supply chain	Ensure agriculture and food supply chains trust and traceability	Conceptual/ implementation
(Gonczol et al., 2020)	Blockchain in supply chains	Blockchain challenges and benefits review and summarisation in supply chain	Survey/ Review
(Centobelli et al., 2021)	Circular supply chain	Fill the gap in circular supply chains traceability, transparency, and trust	Conceptual Framework
(Kouhizadeh & Sarkis, 2018)	Greening supply chains	Outlines blockchain application in greening supply chains	Viewpoint Paper/ Insight
(Jraisat et al., 2022)	Reverse supply chain integration	Blockchain role exploration in sustainability performance improvement in reverse supply	Qualitative Research
(Chang & Chen, 2020)	Blockchain in supply chain management	Study blockchain current development and potential implementations in supply chain management	Systematic Literature Review

c. Gap in Knowledge

Gap in knowledge refers to the insight or understanding absence regarding a specific component of subject that demands deeper investigation. The importance of coming up with the knowledge gaps helps the academics and researchers in determining the domain that needs further exploration, examination, or study to enhance the understanding and widen the robustness of the current comprehension. As seen from the above table, multiple authors investigated both usage of Large Language Models (LLMs) and Blockchain in supply chain management in various methodologies and techniques over the years to understand how to best implement both technologies to enhance supply chain processes. For example,

But current studies specifically only discuss the wide implementations and incorporation without particularly focusing on customer review analysis optimisation in reverse supply chains (Ng et al., 2023). The existing research overlook how sophisticated sentiment analysis, can significantly filter the returns and recalls management by determining product issues from customer reviews. The targeted review analysis potentially streamlines processes for determining specific part of a certain review instead of sending the entire customer feedback and direct these insights to the suitable stakeholders more efficiently in reverse logistics. Also, blockchain incorporation can better improve the process by ensuring the customer review traceability and authenticity that will maintain decentralised systems data integrity. Yet, A wide gap in blockchain ability to support AI study that manage the issues of scalability by only distributing the relevant customer insights instead of duplicating entire feedback, hence, reverse supply chain operations response time and overhead can be reduce (Kouhizadeh & Sarkis, 2018, Jraisat et al., 2022).

d. Thesis Contribution

3. Methodology

3.1 Qualitative Analysis and Data Collection (add paper reference for each section stating type, what has been done, and by whom)

Supply Chain Blockchain Inputs

The supply chain data provided in the data serves as a blueprint to understand how each of the components functions within a blockchain-advanced supply chain management system, complemented by AI-driven sentiment analysis. Every column in the dataset has its importance and focuses on tracking and monitoring the product, order, and financial transaction flow between the critical entities, namely customers, retailers, warehouses, and suppliers. I would provide details on the importance of every column as we go further, with a need to define what it plays in the modern-day efficient supply chain.

To begin with, the foundation of the data consist of the Customer Order Date (cust_ordDate), Customer Due Date (cust_dueDate), and Month (month). The "cust_ordDate" begins this supply chain journey. It is the time when the customer places an order for certain goods, thereby launching a series of following actions, including order processing, delivery scheduling, and payment effecting. In a blockchain-integrated system, this date is recorded immutably to provide an identical and transparent point of reference for all parties. This, in turn, reduces disputes and reliably meets customer expectations. Next, "cust_duedate" supposed to indicate the approximate time of delivery for an order. This has direct repercussions for customer satisfaction and loyalty, one needs to be very careful about setting appropriate expectations among clients. Due dates can be reinforced by the power of blockchain through smart contracts by making sure that at all times when conditions warrant action- such as the dispatch of a product or notifications in case of delays-automation in the supply chain acts in a timely and responsive manner. The "month" column extracts or records the month of the date of the order, providing insights into seasonal trends and demand patterns. This will also enable the

business to predict when peak periods will strike and thereby work out an inventory strategy to avoid the perils of overstocking or stockouts. Such foresight will be essential in ensuring that the supply chain functions optimally.

Then, the data stores customer interactions in the Customer ID (cust_id), Customer Request ID (cust_req_id), Product Name (prod_name), Customer Quantity Request (cust_quan_req), Customer Price Order (cust_price_ord), and Customer Fulfillment (cust_fulfilled) columns. The “cust_id” is a unique identifier of the customer through which orders and interactions are tracked at an individual level. In blockchain, each transaction that will be tagged to this ID will be recorded in a secure way, enhancing data integrity and accountability. This, in turn, provides much-needed transparency during disputes, like returning goods or complaints of any sort, wherein a traceable history line can speed up the resolution process. Similarly, the “cust_req_id” uniquely identifies each request made by a customer so that the supply chain can track multiple orders from the same customer without mixing them up. That is pretty fine-grained, and complex supply chains need this so that every order is processed just right. It also makes it easier to analyse customer satisfaction in relation to each request, thus obtaining insight into preferences and areas where there is probably most room for improvement. The “prod_name” column lists the specific product ordered, an essential attribute that is necessarily required for inventory and order management purposes. On any blockchain-enabled system, this would mean every product will be tokenised with its unique digital identity. This ensures complete traceability, reducing all possibilities of errors or fraudulent cases to ensure correct product delivery each time. “Cust_quan_req” is the quantity ordered by the customer, which is one of the most important stock factors. Accurate recording thus ensures the correct execution of orders taken and the maintenance of stock. Blockchain technology coupled with real-time sharing of data allows all stakeholders in the tracking and responding to stock levels to ensure that there is not a single case of shortage or excess that reoccurs. The column “cust_price_ord” captures the price at which the customer ordered the product. This information is crucial and needs to be made available for financial transparency, first and foremost, for the purpose of ensuring that bills and their respective payments are correctly issued. On a blockchain system, these prices get recorded within an immutable ledger, forbidding unauthorised changes and hence cultivating trust among interacting parties. Last, the “cust_fulfilled” is the binary indicator for if the customer's order is fulfilled. The column is indispensable for tracking order statuses and making sure deliveries are timely. In blockchain-integrated systems, the status for the fulfillment can be automatically updated by smart contracts themselves, reducing human error and improving the efficiency of processing.

Subsequently, the backbone of order processing, retailer data, is contained in the Retailer ID (ret_id), Retailer Price Offer (ret_price_off), Retailer Quantity Available (ret_quan_avail), and Retailer Available Difference (ret_avail_diff). The “ret_id” identifies the retailer who actually fills the order. As noted, this ID plays an important part in tracking accountability through every level of the supply chain. On the blockchain, this ID is attached to each related transaction, ensuring that every role played by each retailer is transparently and traceably captured. “ret_price_off” stands for Retailer Price Offered, the price that a retailer would give out to sell a particular product. This might be different from what the customer pays since there can be many discounts or other considerations. Price transparency avoids disputes, and with blockchain, these records remain tamper-proof, ensuring that all fair and trustful deals occur. “Ret_quan_avail” is the quantity available with retailer is of prime importance to place orders and deliver value to the customers. In real life, this needs to be updated in real time and integrated with blockchain for better visibility throughout supply chain. While the “ret_avail_diff” is the difference between what customer requests and what is in stock. This metric identifies where there is likely to be a problem with stocks early so the retailer can actively correct the stock levels, or

adjust the customer expectation of stock. Blockchain can automatically trigger an action from this data to ensure supply can match demand.

Afterwards, more data like Warehouse ID (whs_id), Warehouse Price Offer (whs_price_off), Warehouse Quantity Available (whs_quan_avail), Warehouse Available Difference (whs_avail_diff), Warehouse Price Order (whs_price_ord), and Warehouse Fulfillment (whs_fulfill), from the warehouse, and Supplier ID (spl_id), Supplier Price Offer (spl_price_off), Supplier Quantity Available (Supplier Quantity Available), Supplier Availability Difference (spl_avail_diff), Supplier Fulfillment (spl_fulfill), were also stored to ensure seamless operations. Initially, the “whs_id” denotes the unique identifier for the warehouse storing the product. This will be important because it will enable physical tracking of the goods. Through Blockchain, this information becomes available for all parties that are involved in this chain of value, enabling better coordination and reduction of costly mistakes in the management of inventories. Then, the “whs_price_off” refers to the price the warehouse offers to the retailers or suppliers. Such transactions have to be transparent, and blockchain keeps these records in such a way that it builds trust and minimises the chances of disputes. Subsequently, “whs_quan_avail” is the quantity on-hand at the warehouse. This is one of those very important issues in any industry dealing with products since proper management of the inventory level is needed so that supply may meet demand. Blockchain and real-time updates prevent discrepancies; hence, smooth operations are ensured. Next, “whs_avail_diff” The difference between requested and available at the warehouse level, useful for pinpointing any bottlenecks. Using blockchain, automation in response to these kinds of discrepancies will keep moving goods within the supply chain. After that, “whs_price_ord” is the price at which the warehouse procures the goods. This gives one information for financial planning. By blockchain, those records are transparent and secure; hence, correct cost management is easily achieved with few or no disputes. The “whs_fulfill” Indicates whether the warehouse has fulfilled its part in an order. This status is of utmost importance, and with blockchain, it is able to automate updates for efficiency in operations.

Moving on to supplier data, “spl_id” uniquely identifies the supplier, allowing tracking of goods to their source. Since blockchain provides a trace for goods with a sense of responsibility, this ID attaches to all transaction records related to the particular material/goods. “Spl_price_off” contains the price that a supplier is offering for goods. Equitable pricing of goods through the value chain requires openness in pricing by suppliers. Blockchain ensures that the record of the goods' prices by suppliers is duly kept in security. Then, the “spl_quan_avail” reflects the amount of the stock available with the supplier. This is instrumental in making sure the supplies meet the demand of the products. Blockchain and hence updates in real-time helps to keep the exact amount of inventories on hand to avoid stockouts. Also, the “spl_avail_diff” showcased the difference in stock levels between what is on request and what is available at the supplier's end. This helps in pre-emptive measures against a shortage in the supply. Blockchain, in this regard, will trigger an automated response as and when required to restock the warehouses. While the “spl_fulfill” is the actual order by the supplier, which will make sure the supply chain functions perfectly; today, using blockchain and smart contracts, the same thing can be achieved without unnecessary delay to maintain efficiency.

Lastly, the data also accommodate the financial transactions data like to ensure accountability and transparency. Retailer Paid (ret_paid) that flag the situation of payment by the retailer for goods, which is one of the major driving forces to maintain fiscal transparency. Blockchain will ensure that the book of payments remains safe from dispute and all dues are paid. Retailer Delivery (ret_dlv) shows delivery status whether the retailer has delivered the merchandise to the subsequent stage in the supply chain. On-time delivery is critical to satisfying customers. Blockchain automates status updates for efficiency in the process. Warehouse Paid (whs_paid) records the warehouse payments,

which is an important aspect of financial management. Blockchain securely records these transactions to ensure fair compensation among parties. Warehouse Delivery (whs_dlv) presented whether or not the warehouse has delivered the goods - this is a critical status that tracks the movement of goods. Blockchain automates this process and can reduce such delays. Supplier Paid (spl_paid) reflects payments to suppliers. It needs to be timely, especially on the supplier's end. Blockchain secures the records of such transactions, engenders trust, and reduces the chances of disputes. Supplier Delivery (spl_dlv) represents whether the supplier has delivered the goods. This is towards tracking the progress within the supply chain. Updates such as this are automated under blockchain: goods are delivered to that effect. Customer Paid (cust_paid) is a status variable to track if the customer paid for goods; a very important aspect of cash management: timely payments are of essence in maintaining fluidity of finances, and through blockchain, the interface ensures the security and transparency of the transactions.

Each column in the supply chain dataset has an important role and contribution to the management of complex processes in a blockchain-enhanced supply chain. The supply chain could be more effective and transparent concerning the recording and tracking of orders, pricing, inventory, fulfillment, and payment with precision. All these processes are further empowered through blockchain technology that records all transactions on its immutable ledger for truthfulness of data access to every party. This engenders trust and leads to a better client satisfaction and efficient supply chain management.

Supply Chain Sentimen Analysis Inputs

The provided analysed dataset is drawn from various sources available on Kaggle. Each one contributes a different valuable piece of information about Amazon customer reviews. It contains several datasets compiled by various contributors, with a focus on aspects related to review data, sentiment analysis, and product feedback. It is a contributed dataset by Kritanjali Jain, and the reason it is comprehensive is that it contains reviews for a wide range of products on Amazon and hence covers a broad base for sentiment analysis regarding customer feedback. Equally useful is the dataset contributed by Bittlingmayer, as it is tailor-made for sentiment analysis. Besides the vast data it contains, one can also use it to train the models for understanding the sentiment of customers. Abdallah Wagih's dataset, while adding to the richness of diversity and putting a strong emphasis on textual review content, is rather useful for natural language processing tasks. On the other hand, the Datafiniti dataset deals with customer reviews of Amazon products-invaluable every time there appears the need to analyse customer opinion and track the performance of each and every product in time. Last but not least, this dataset contributed by Yasserh is really huge in the aspect of Amazon product reviews; it has very comprehensive feedback for so many categories, which forms a really strong basis for comprehensive sentiment analyses.

Each of these diverse datasets, focused on their own aspect, in turn creates an enormous library of customer reviews that detail the landscape of Amazon's e-commerce ecosystem. The consolidated dataset focuses on two key columns: "Review" and "Sentiment". This allows the dataset to be streamlined in focus on focused sentiment analysis while avoiding excessive complexity that may undermine efficiency and appropriateness.

The "Review" column carries the actual text of the customer reviews constituting the substance of this data set. They can be short and concise, or they can be as descriptive. Much information about the experience that customers have gone through with various products can be derived from these reviews. This column is of immense value with regard to natural language processing tasks such as sentiment analysis, keyword extraction, and thematic analysis. This provides a direct understanding

of what was liked about the product, what was disliked, and even their general satisfaction. Text analysis of such nature is done for businesses in order to have more in-depth information about customer satisfaction, performance of a product, and where improvements might be called for. Furthermore, insights could help in product development, marketing strategies, and help improve customer service.

Sentiment categorises the emotional feel of these reviews into three major categories: "Positive", "Negative", and "Neutral". In some sense, this makes it easier to look into the overall tone in customer feedback without having to go through each review. The information here usually is pulled from pre-labeled datasets based on the sentiment expressed within the reviews. The tagging is done in such a way that, for instance, reviews that express satisfaction about a product are tagged as "Positive", while those that express dissatisfaction are tagged as "Negative", and those reviews that do not give a strong emotional feeling are tagged as "Neutral." The sentiment data gives the very crucial basis for sentiment analysis, whose end result is to get the general emotional response of customers toward a product. This would help businesses in monitoring the level of customer satisfaction over time, thereby understanding how the level of satisfaction is changing. It may be extended to forecast even the success of a product in the near future based on prevailing customer feedback.

The "Reason Extracted" is done for a reason: to reduce the data that will be used in sentiment analysis of the "Review" and "Sentiment" columns. The sentiments of the customers will be focused on a large array of products. Such narrow focus shall allow for more relevance and efficiency in the analysis; hence, it will be relevant to the businesses using such data to arrive at informed decisions.

Thus, the successive steps of processing this data involve actually applying several steps of NLP techniques to review text, such as tokenisation, stopword removal, stemming, or lemmatization, to prepare the text for analysis. It follows that now, one can apply sentiment analysis algorithms on this text, with advanced state-of-the-art LLM models such as RoBERTa. In other words, the Sentiment column will be our target variable for this kind of analysis, enabling us to train models to predict sentiment given some review text. This would allow in-depth understanding of customer opinions from which insights might be obtained and inform business decisions on the basis of product feature improvements, marketing strategy refinements, or enhancements in customer service.

This dataset constitutes an amalgamation of comprehensive Amazon review datasets, presented as a dataset with "Review" and "Sentiment" columns constituting the very basis for detailed sentiment analyses. The concentration on these very columns alone assures the optimality of the dataset in light of further processing for efficiency and insight into customer sentiments across a wide array of products. Such an analysis would be very useful in providing an insight into the behaviour of customers and the performance of various products, hence allowing businesses to work out improvements in their offerings to meet the needs of their customers. The various sources from which the data are drawn ensure a broad outlook and encompass sentiments from a vast array of product categories and a wide range of customer experiences. This will further enable the processing and analysis to be done efficiently for a better understanding of the e-commerce landscape, hence helping businesses make informed decisions based on real customers' feedback.

3.2 Quantitative Analysis (a paragraph)

In this section we will use blockchain for {}, and sentiment analysis for {}

3.3 LLM Enabled Supply Chain Blockchain Framework

The figure below showcased a complex supply chain information sharing system powered by blockchain, smart contracts, and natural language processing. Fundamentally, the purpose of the system is to help each component in the supply chain to communicate with each other by providing transparency and traceability to fulfill customer order and real-time customer feedback integrated into supply chain management processes toward an efficient, transparent, and responsive supply chain process.

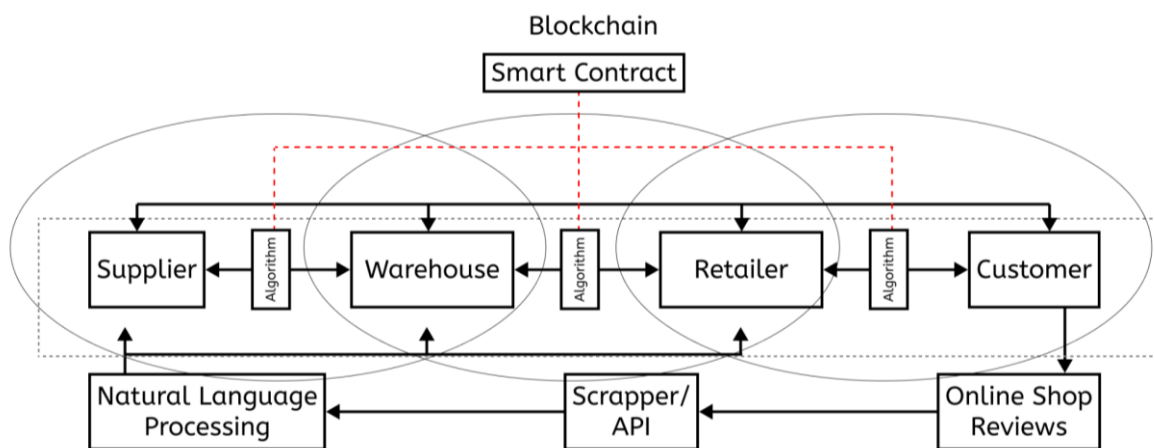


Figure 1. Supply Chain Blockchain Smart Contract Diagram

The diagram is framed around the major components that make a supply chain: the suppliers, warehouses, retailers, and customers. All these components are linked by the blockchain algorithms that assist in information, goods flow, and communication from one to the other. Through the blockchain algorithms, it is possible to have sophisticated data processing integrated into the various functions across the chain.

In the most basic form, the system is dependent upon data sharing between components, which will enable them to better understand each other condition and better anticipate the upcoming demand trend through the ease of communication. Also, the provided data will go through analysis with the ability to provide insights on how to make decisions. This analytic layer takes source raw materials and nominates actionable knowledge derived from a variety of sources.

Another critical feature of the system is the ability to leverage customer reviews of the online shop. The reviews are collected with the help of a scraper or API that can effectively extract big volumes of text data. These processed, informative data will then be processed by the Natural Language Processing (NLP) techniques of the supply chain, helping and guiding operational decisions. This way, it will help to understand the sentiment or opinion that the customer alludes to in their review.

An overlay of blockchain and smart contracts across the supply chain network symbolizes the system capacity to enable complete integration of the blockchain technology feature. This lays across a decentralized, immutable ledger for all transactions across the supply chain. Therefore, this kind of ledger has high transparency and robust security. The automated data processing of preset rules is the defining feature of smart contracts, which eliminates most of the processes involved in the manual

processing stage. As a result of the elimination of intermediaries, such contracts are efficient and reduce the possibility of human error or fraud.

The data, in conjunction with the blockchain, functions between the supplier, warehouse, retailer, and customer. Suppose a customer is posting a review on a product, the information would be collected and pass through NLP algorithms and get classified based on the sentiment (positive, neutral, negative) and category (retailer, warehouse, supplier). Insights from it would be pushed to the corresponding supply chain stakeholders in the form of smart contracts. Therefore, suppliers and warehouses will be empowered to re-schedule operations on a real-time basis and have the inventory level close to what the customer has demanded and expects.

Now, with such a dynamic and sensitive ecosystem of the supply chain as a figure where each supply chain component can easily communicate and assess each other's condition, real-time customer feedback through NLP, backed by blockchain technology, and operated with smart contracts ultimately leads to nothing less than improvement relentlessly and works operationally to excel. It not only harmonizes the members of the supply chain through making the process transparent and one of trust but also connects the supply chain further with the needs of customers for their betterment and reduced inefficiency.

The presented diagram is a conceptual model of the "Supply Chain Component Wallet," which would act as the digital repository with the intention to enhance the efficiency and transparency of supply chain operations by integrating blockchain technology. At the basic level, it is segmented into two main layers: Product Attribute Pockets and Agreement Pockets, all birthed under the overarching Supply Chain Component Wallet.

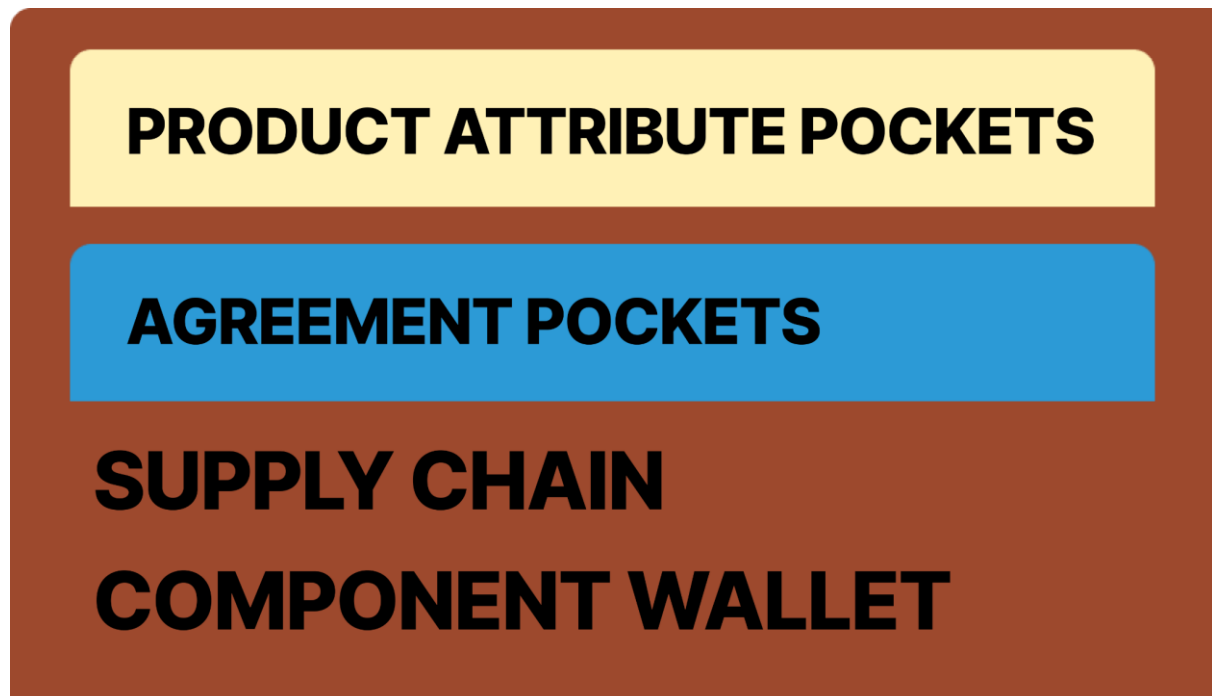


Figure 2. Supply Chain Component Wallet

On top of the structure are the Product Attribute Pockets. This could be visualized as something that should retain all information regarding product attributes from the supply chain. These can be any attribute from size, weight, color, among other product descriptors that happen to be real keys

for any effective inventory tracking or management plan. The system creates specific pockets for specific product attributes, and therefore, all information about each product will be well put together and easily accessible for tracking purposes or retrieval whenever called upon.

The pockets of agreements lie under the pockets of product attributes. This layer could be used for storing the terms and conditions agreed upon by various participants in the supply chain. Agreements can contain a whole host of contractual details pertaining to pricing, delivery schedules, quality standards, or even payment terms. In part, these may be digitized and stored within structured pockets of the agreement. Doing so enables smart contract-based automated enforcement of contract terms. This automation limits the occurrence of disputes and ensures that all parties conform to the decided terms, hence increasing trust and reliability within the supply chain.

This then is the envisioned form of the Supply Chain Component Wallet—facing, a layered model that would go a long way toward bringing in a structure for the management of critical information within a supply chain. Blockchain technology does this much more transparent, secure, highly efficient. The Product Attribute Pockets provide that exact product details are clearly tracked, while Agreement Pockets ensure any contractual obligations are met as the result of smart contracts. Those layers, integrated as a component wallet within a supply chain, compose the building blocks for a very powerful system that supports improved decision-making and allows better collaboration among participants in the supply chain.

Below is an agent-based model diagram that provides the details of the interaction and informational flow within the supply chain network, outlining the integration of Supply Chain Component Wallets across different stages, from goods suppliers to customers. It captures chronologically product inquiries and their responses between participants in the supply chain and puts forth quite clearly the place of digital wallets in facilitating these interactions.

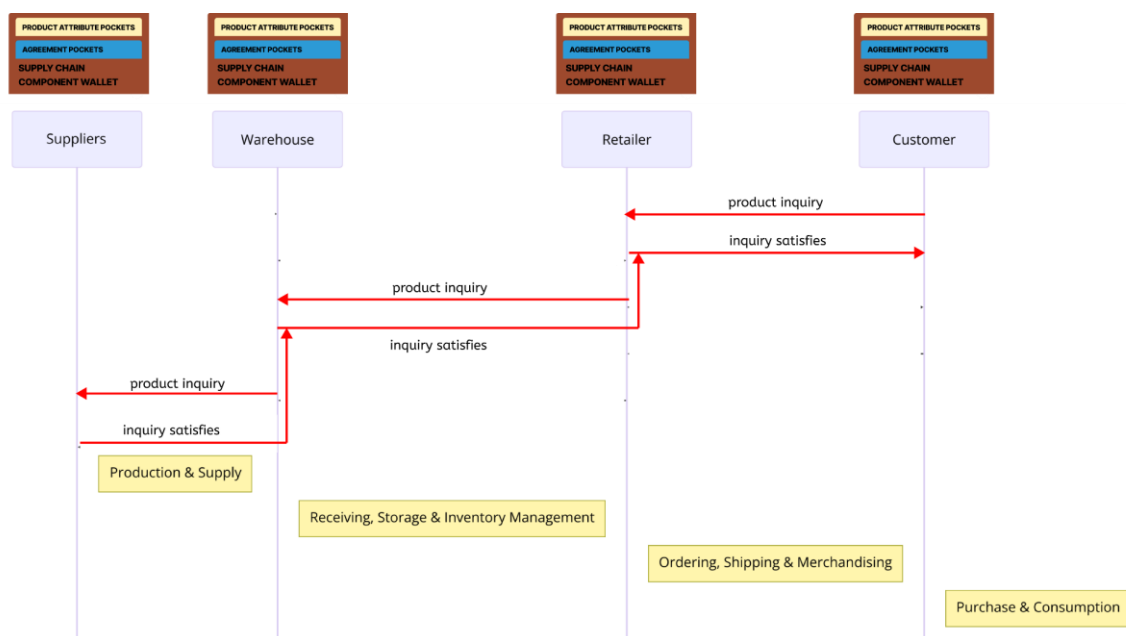


Figure 3. Supply Chain Components Agen-Based Flow

Starting with the Suppliers, the diagram shows that each supplier is fitted with a Wallet of Supply Chain Components with Product Attribute Pockets and Agreement Pockets. The suppliers are charged with the primary role of Production & Supply of goods. On inquiry on a product by any downstream

component, such as a warehouse, responds to the inquiry by opening its wallet for the corresponding product attributes and agreements in an effort to satisfy the inquiry.

The second supply-chain node is the Warehouse, which has also issued its own version of a Supply Chain Component Wallet for dealing with product information and contractual agreements. On the part of the warehouse, there will be receiving, storage, and inventory management. Upon inquiry about certain products by a retailer, it checks with the stored product attributes and agreements to give an answer. Such interaction allows a warehouse to handle its inventory and respond effectively to inquiries.

It is followed by the Retailer in the chain, equipped similarly with a Supply Chain Component Wallet to facilitate Ordering, Shipping & Merchandising. The retailer initiates product inquiries to the warehouse in order to check availability—to make sure it has what the customer wants. After receiving product inquiry from a customer, it goes into its wallet, details about the product and confirms the details of his order. This will facilitate a seamless flow of information for retailers to manage stock and fulfill orders efficiently.

At the other end of the supply chain is the Customer, whose primary activities are Purchase & Consumption. On product inquiries by customers to gather information for the purpose of making a purchase decision, retailers can quickly respond to such inquiries as comprehensive data is available with them in their wallets and provide accurate and update product information to the customer.

It shows the product inquiries and then, subsequently, the satisfaction of those inquiries by the red arrows. This bidirectional flow of information brings out the dynamic nature of a supply chain in which any component not only sends but also responds to requests, based on data held within their digital wallets. Supply Chain Component Wallets: All product attributes and agreements are kept in digital wallets, ensuring very smooth and efficient interaction among them.

The diagram shows a very integrated supply chain system where the use of digital wallets ensures transparency, efficiency, and response throughout. The structured inquiry and the response of the product indicate that real-time data access and management, which blockchain technology enables, are important for the optimization of supply chain operations in order to realize customer satisfaction.

The next detailed diagram explains the flow of transactions or activities and data in a blockchain-enabled supply chain system. It involves customers, retailers, warehouses, and suppliers as different involved entities that engage through one central entity called the Transaction Database. Digital wallets are individual, with each of the transacting parties securely managing and storing their own transactions data.

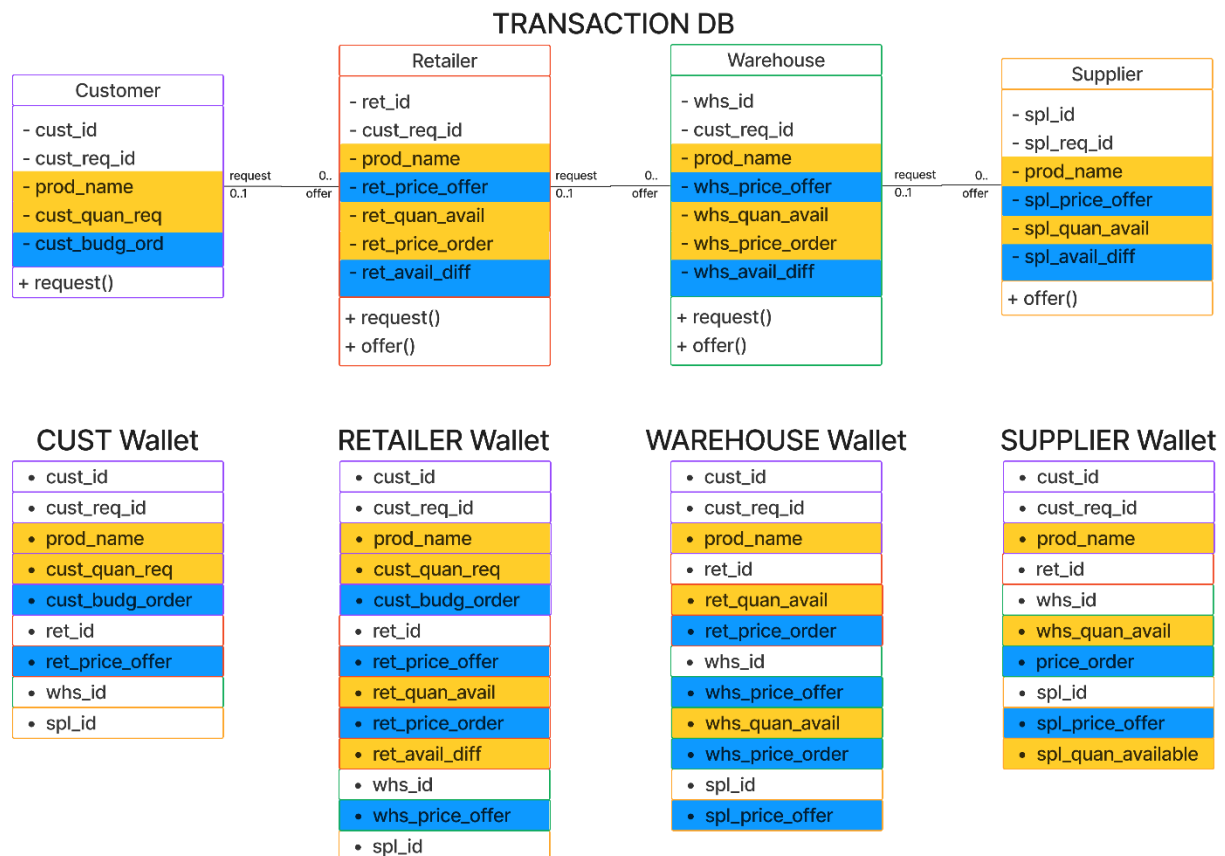


Figure 4. Transaction Database

This system will maintain a Transaction Database that serves as the root of the chain, offering one central location for all types of transaction information. The communication among the different supply chain entities is enhanced in this way because data are exchanged effortlessly. Every participant keeps copies of the essential attributes in an absolutely transparent and traceable way for every single transaction.

It has fields for customer ID (cust_id), customer request ID (cust_req_id), product name, quantity requested to the retailer, and budget order. Customers generate requests for products, after which retailers process the request and note it in Transaction DB. Customer requests are passed on to the retailers.

The database also contains attributes like retailer ID, customer request ID, product name, retailer price offer, quantity available, price order, and availability difference. The retailers respond to the customer's inquiry and forward it to the warehouses. They handle requests as well as offers quite efficiently with regard to inventory and pricing data in order to meet demanding customers.

Several attributes captured by the warehouse section include warehouse ID, customer request ID, product name, warehouse price offer, availability quantity, price order, and difference of availability. The inventory level is maintained by the warehouses themselves so that they are able to cover retailers' inquiries on the products.

On the other hand, the suppliers section contains supplier ID, customer request ID, product name, price offered by the supplier, quantity available and availability difference. The suppliers provide products to both warehouses and retailers with very important data on availability and their pricing.

In respect, every single element of the supply chain is represented digitally in terms of transactions with their respective data. Accordingly, the Customer Wallet includes, among other things, customer ID, request ID, product name, quantity requested, budget order, retailer ID, and price offer. This wallet keeps all the data of the customers secure yet accessible.

Similarly, the same information like customer ID, request ID, name of the product, quantity available, price offer, price order, warehouse ID, and supplier ID, will reside in the retailer wallet. This helps in managing retailer-specific transactions with its inventory.

Furthermore, the warehouse wallet also contains information about the customer ID, request ID, product name, available quantity, offer price, order price, retailer ID, and supplier ID. This wallet is useful even in tracking the entire operation of the warehouses and their inventory levels at large.

Subsequently, the supplier wallet also contains customer ID, request id, name of the product, price offer, available quantity, retailer's ID, warehouse ID. The wallet helps a lot to suppliers to maintain a record regarding all their offerings and against queries by warehouses or retailers.

The arrows in the top diagram depict the flow of interaction between these entities—namely, a product inquiry and its response sequence. Customers raise product requests to the retailers, who further raise inquiries to the warehouses and suppliers in order to respond to these requests. Each one of them responds in terms of offers created as per the data stored in their respective wallets to make a seamless and transparent transaction process.

The diagram briefly describes a very robust supply chain system and clearly points out how blockchain technology will be capable of making improvements in relation to transparency, efficiency, and security. All data related to the transactions will be stored on the transaction DB and digital wallets individually, which makes every entity able to manage and retrieve information safely. With this system, real-time data sharing, automation in process execution, and optimization of supply chain operations will be realized to ensure customer satisfaction.

Figure 5. Schematic overview of tokenisation in some blockchain enabled supply chain systems, including component-smart contract-customer wallet interactions. Here, tokenisation refers to the creation of digital tokens that correspond to physical products in a supply chain.

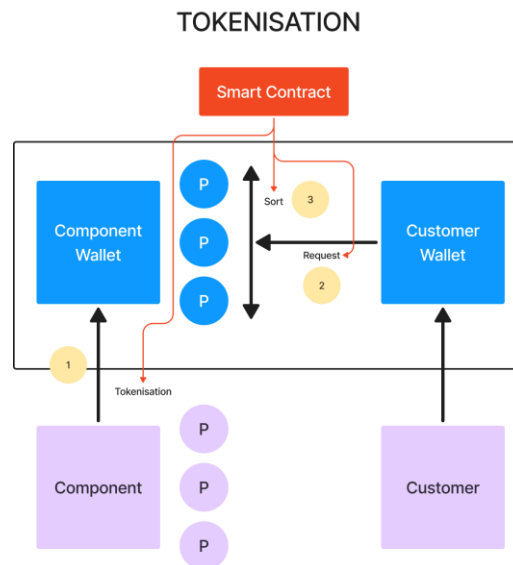


Figure 5. Tokenisation Process in a Blockchain-Integrated Supply Chain Using Smart Contracts

Most of the time, the tokenisation process started by the suppliers or manufacturers. Every other product from then on will be converted into a token and referred to as "P," which in turn will be stored on their Component Wallet. In this regard, tokenisation would ensure each product has its very own unique digital identifier for tracking and management purposes along the supply chain. Subsequently, these tokens would be placed in pools and would lie there awaiting subsequent transactions and processing.

When tokenised, those goods interact with a Smart Contract, which makes part of the management and automation of integral rules and processes directly related to the given items. Smart contracts are practically peculiar self-executing contracts by lines of code inside them, which clearly offer what the terms of the deal are. In more layman terms, transactions become lucid, secure, and automatically executed without any form of intermediaries.

When customers make a request for some product, they do it to the smart contract (step 2). The customer defines what they need, and the smart contract processes this request. It simply means that the smart contract is going to go through all the available tokens of products which are kept in the Component Wallet, checking their criteria about predefined factors such as availability, specification of the product, and so on, to find the most suit that answers the customer's request.

On sorting and identification of the right token, it permits the sell-off of the selected product token to the Customer Wallet. Customer Wallet acts as a sort of secure digital repository in which Product Tokens acquired by the Customer are stored. This allows the customers to access a verifiable, immutable record for purchased products, hence increasing faith and transparency into the Transaction.

This interaction between Component Wallet, Smart Contract, and Customer Wallet shows the efficiency and the security brought about by blockchain technology in the management

of supply chains. By digitising the products into tokens and using smart contracts to help auto-pilot and execute transactions, manual intervention is drastically reduced; minimal errors therefore take place in what is transparently traceable in all transactions.

In other words, the diagram depicts how tokenisation fits into blockchain to drive frictionless and secure operations in the supply chain. Thus, a structure is set in place with the interaction of Component Wallets storing the tokens, smart contracts executing an automated process, and Customer Wallets maintaining secure records that will introduce efficiency, transparency and dependability in supply chain transactions. That way, supply chain participants and end customers can derive greater confidence through building operational efficiencies.

The diagram depicts details of a transaction occurring in the blockchain-enabled supply chain system and outlines the responsibility of smart contracts at different supply chain participants and their associated digital wallets. This links the suppliers, warehouses, retailers, and customers through the blockchain for security and an effective means of carrying out transactions.

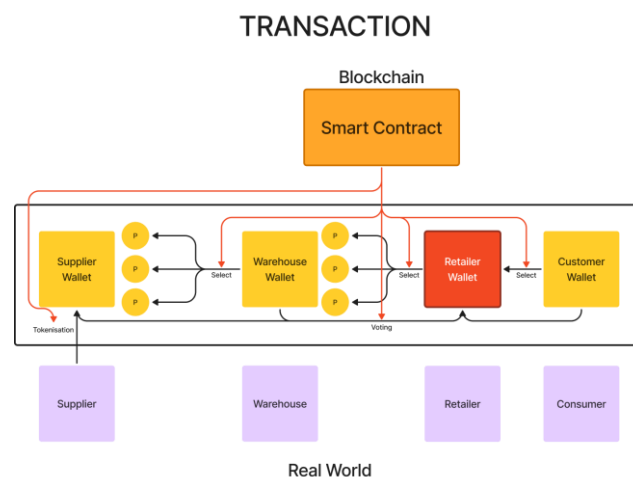


Figure 6. Blockchain-Enabled Supply Chain Transactions Using Smart Contracts and Digital Wallets Illustration

First, the suppliers tokenise their products, meaning that every product is turned into some digital form of a token and kept in the Supplier Wallet. To make the way from production to the end, all products get digital identification so it is possible to track their path from the manufacturing process through delivery. To replenish the inventory, the warehouse has inbound queries to the supplier wallet, which consequently provides tokens needed for restocking. Through this interaction, the smart contract is able to verify and clear the transfer of tokens, therefore ensuring that the warehouse gets a confirmed digital record for goods received.

Similarly, when a retailer needs to stock products to eventually be able to fulfill customer orders, it requests tokens from the Warehouse Wallet. Again, the smart contract guarantees that all of this transaction is properly automated and authenticated. Upon approval, the

promised amount of product tokens is then transferred into the Retailer Wallet, returning some form of digital inventory count so that the retailer accounts properly for what stock they have at hand. This obviously needs to make sure that retailers are located whereby they can handle stocking with efficiency and proper responses to consumer demand.

Lastly, in the transaction process is a customer making a purchase from a retailer. A request is raised by the retailer for a transaction. Post request, the smart contract processes and validates the request. The transaction is now confirmed. Product tokens are transferred from the Retailer Wallet to the Customer Wallet, providing customers with verifiable digital proof of what they purchased—amplifying transparency and consequently creating more trust in transactions.

Such transparency and security are managed by the smart contract in every transaction along the process. Since it ensures that all conditions are met as required, the transfer of tokens is never permitted, thereby preserving the integrity of the supply chain. Digital wallets, at every stage, perform in a manner much like secure storage, product tokens and records of transactions, thus making it efficient and traceable for goods consigned from suppliers to customers.

The diagram illustrates how blockchain technology and smart contracts feature to handle each process of transaction in a supply chain very effectively. The system makes itself transparent, reliable, and secure, not only effective from producers but also from warehouses, retailers to customers, through the technologizing of products and the way of their use in companies. This not only advances the traceability and reliability of the transactions but can also further build more on trust between the members involved in the supply chain. Then, this enables a blockchain system to make a supply chain easier to be manipulated and vigorous with regard to meeting customer needs more effectively and efficiently.

3.3 Research Heuristic Code

Smart Contract Abbreviations

Variable	Abbreviation	Description
cust_ordDate	Customer Order Date	The date when the customer placed the order.
cust_duedate	Customer Due Date	The due date for the customer order.
month	Month	The month part of the order date.
cust_id	Customer ID	A unique identifier for the customer.
cust_req_id	Customer Request ID	A unique identifier for the customer's request.
prod_name	Product Name	The name of the product ordered by the customer.
cust_quan_req	Customer Quantity Requested	The quantity of the product requested by the customer.
cust_price_ord	Customer Price Ordered	The price of the product ordered by the customer.
cust_fulfill	Customer Fulfillment	Indicates whether the customer's order has been fulfilled.

ret_id	Retailer ID	A unique identifier for the retailer.
ret_price_off	Retailer Price Offered	The price offered by the retailer.
ret_quan_avail	Retailer Quantity Available	The quantity of the product available with the retailer.
ret_avail_diff	Retailer Availability Difference	The difference between requested and available retailer quantities.
ret_price_ord	Retailer Price Ordered	The price at which the retailer ordered the product.
ret_fulfill	Retailer Fulfillment	Indicates whether the retailer has fulfilled the order.
whs_id	Warehouse ID	A unique identifier for the warehouse.
whs_price_off	Warehouse Price Offered	The price offered by the warehouse.
whs_quan_avail	Warehouse Quantity Available	The quantity of the product available at the warehouse.
whs_avail_diff	Warehouse Availability Difference	The difference between requested and available warehouse quantities.
whs_price_ord	Warehouse Price Ordered	The price at which the warehouse ordered the product.
whs_fulfill	Warehouse Fulfillment	Indicates whether the warehouse has fulfilled the order.
spl_id	Supplier ID	A unique identifier for the supplier.
spl_price_off	Supplier Price Offered	The price offered by the supplier.
spl_quan_avail	Supplier Quantity Available	The quantity of the product available with the supplier.
spl_avail_diff	Supplier Availability Difference	The difference between requested and available supplier quantities.
spl_fulfill	Supplier Fulfillment	Indicates whether the supplier has fulfilled the order.
ret_paid	Retailer Paid	Indicates whether the retailer has been paid.
ret_dlv	Retailer Delivery	Indicates whether the retailer has delivered the product.
whs_paid	Warehouse Paid	Indicates whether the warehouse has been paid.
whs_dlv	Warehouse Delivery	Indicates whether the warehouse has delivered the product.
spl_paid	Supplier Paid	Indicates whether the supplier has been paid.
spl_dlv	Supplier Delivery	Indicates whether the supplier has delivered the product.
cust_paid	Customer Paid	Indicates whether the customer has paid for the order.
cust_dlv	Customer Delivery	Indicates whether the product has been delivered to the customer.
IRN	Internal Reference Number	A reference number is used internally in the algorithm.
IC	Internal Condition	A condition used internally in the algorithm.
PROD	Product	The product involved in the transaction.
PC	Product Condition	A condition related to the product used in the algorithm.
DLVR	Delivery	Indicates the delivery status.
DC	Delivery Condition	A condition related to delivery is used in the algorithm.
DD	Due Date	The due date for the current stage in the timeline.
ED	Event Date	The event date for the current stage in the timeline.

Smart Contract Algorithm

DELIVERY & TRANSACTION RECORDING FUNCTION

STEP 1: Define the function `MARK_LIGHT` with the parameter `TIMELINE`.

STEP 2: Create four columns using `ST.COLUMNS`.

STEP 3: Iterate through each stage in the timeline:

STEP 3.1: CUSTOMER FUNCTION

IF `size` is provided:

GENERATE customer order dates, month, customer IDs, request IDs, product names, quantities, prices, and due dates.

RETURN a DataFrame with the following columns: `cust_ordDate`, `month`, `cust_id`, `cust_req_id`, `prod_name`, `cust_quan_req`, `cust_price_ord`, `cust_fulfill`, `cust_duedate`.

STEP 3.2: RETAILER FUNCTION

IF `size` and `cust` DataFrame are provided:

GENERATE retailer IDs and customer request IDs.

MERGE retailer data with the customer data.

CALCULATE retailer offer prices, available quantities, availability difference, and order prices.

DETERMINE fulfillment status based on availability.

RETURN the merged DataFrame with additional retailer columns: `ret_id`, `ret_price_off`, `ret_quan_avail`, `ret_avail_diff`, `ret_price_ord`, `ret_fulfill`.

STEP 3.3: WAREHOUSE FUNCTION

IF `size` and `ret` DataFrame are provided:

GENERATE warehouse IDs and customer request IDs.

MERGE warehouse data with the retailer data.

CALCULATE warehouse offer prices, available quantities, availability difference, and order prices.

DETERMINE fulfillment status based on availability.

RETURN the merged DataFrame with additional warehouse columns: `whs_id`, `whs_price_off`, `whs_quan_avail`, `whs_avail_diff`, `whs_price_ord`, `whs_fulfill`.

STEP 3.4: SUPPLIER FUNCTION

IF `size` and `whs` DataFrame are provided:

GENERATE supplier IDs and customer request IDs.

MERGE supplier data with the warehouse data.

CALCULATE supplier offer prices, available quantities, and availability difference.

DETERMINE fulfillment status based on availability.

RETURN the merged DataFrame with additional supplier columns: `spl_id`, `spl_price_off`, `spl_quan_avail`, `spl_avail_diff`, `spl_fulfill`.

STEP 4: Display the status in the respective column.

IF `DD >= ED`:

IF `IRN == TRUE`: THEN `IC = "GREEN"`

ELSE: `**IC = "GRAY"`

IF `PROD == TRUE`: THEN `PC = "GREEN"`

ELSE: `**PC = "GRAY"`

IF `DLVR == TRUE`: THEN `DC = "GREEN"`

ELSE: `**DC = "GRAY"`

ELSE: SET `IC`, `PC`, `DC` to `"RED"`

STEP 5: Calculate the due date and event date, converting them to date format if needed.

STEP 6: Define the function `data_gen`:

STEP 6.1: Initialize `ret_paid_gen`, `ret_dlv_gen`, `whs_paid_gen`, `whs_dlv_gen`, `spl_paid_gen`, `spl_dlv_gen` as `False`.

STEP 6.2: Loop through the dataset:

IF conditions for fulfillment and payment are met:

UPDATE the statuses for `cust_paid`, `ret_paid`, `whs_paid`, `spl_paid`, and their delivery statuses.

GENERATE a new record with updated statuses.

IF all statuses are fulfilled: **RESET** the index and flags for the next iteration.

RETURN the updated dataset after merging with transaction records.

STEP 7: Define the function `dat_merge`:

STEP 7.1: Merge the generated data with the transaction records.

STEP 7.2: Apply conditions to update fulfillment and delivery statuses.

RETURN the updated DataFrame.

STEP 8: Define the function `dframe`:

STEP 8.1: Sort and group data by various criteria:

SORT by `ret_quan_avail`, `ret_price_off`, `whs_quan_avail`, `whs_price_off`, `spl_quan_avail`, and `spl_price_off`.

GROUP by `cust_id`.

STEP 8.2: Generate the final data with and without backorders.

RETURN two DataFrames: `data_wbc` and `data_wobc`.

The Delivery & Transaction Recording Function is conceptualised as a smart contract designed to automate and secure the order fulfillment process across a blockchain-based supply chain. This algorithm methodically tracks the flow of goods and payments from the moment a customer places an order until the supplier completes the delivery, ensuring that all participants—customers, retailers, warehouses, and suppliers—fulfill their roles according to predefined conditions embedded within the smart contract.

Step 1 initiates the process by defining the ``MARK_LIGHT`` function, which serves as the core mechanism for managing and monitoring the timeline of events in the supply chain. This function uses a ``TIMELINE`` parameter to control and enforce the sequence of operations, such as order placement, payment confirmation, and product delivery. The ``MARK_LIGHT`` function is central to the contract's logic, ensuring that each subsequent step adheres to the established timeline and conditions. Step 2 focuses on establishing four critical columns within the smart contract, each representing the status of a key checkpoint in the supply chain: customer payment, retailer fulfillment, warehouse processing, and supplier delivery. These columns act as state variables that record the real-time status of these checkpoints. For example, the contract tracks whether a customer has completed payment, whether the retailer has fulfilled the order, whether the warehouse has processed the request, and whether the supplier has dispatched the goods. These variables are crucial for maintaining transparency and ensuring that each participant's obligations are met. Step 3 involves iterating through the supply chain timeline, beginning with Step 3.1, where customer order details are generated, including order dates, product names, quantities, and prices. This data forms the foundation for the rest of the process, as it is used to verify the capabilities of the other participants in the supply chain. Step 3.2 assesses the retailer's ability to meet the customer's demands, checking inventory levels and pricing to ensure the retailer can fulfill the order. Step 3.3 shifts focus to the warehouse, evaluating its ability to supply the necessary products to the retailer. Finally, Step 3.4 examines the supplier's capacity to meet the warehouse's needs, ensuring that the products are available for delivery. Each of these steps sequentially ensures the smooth operation of the supply chain by verifying that each participant can fulfill their respective roles. Step 4 updates and displays the status of these checkpoints in the respective columns, using color-coded indicators—"GREEN" for successful completion, "GRAY" for pending or in progress, and "RED" for failures or delays—to provide a clear, real-time overview of the order's progress. This step is essential for maintaining visibility across the supply chain, allowing all participants to monitor the status and ensuring that any issues are quickly identified and addressed. Step 5 focuses on calculating and formatting due dates and event dates, which are crucial for ensuring that all actions are completed within the specified timelines. The contract uses these dates to enforce deadlines and may trigger penalties or status updates if conditions are not met. This step ensures that the supply chain operates efficiently and that delays are minimized. Step 6 defines the ``data_gen`` function, which simulates ongoing processes within the supply chain by iteratively updating the status of payments and deliveries as the order progresses. This function reflects real-time changes within the contract, ensuring that the state is consistently updated to match the current status of the supply chain. Step 7 introduces the ``dat_merge`` function, which combines the newly generated data with existing transaction records. This ensures that all updates are synchronized and that the records accurately reflect the current state of the supply chain. The function applies specific conditions to verify that all required actions have been completed before advancing to the next stage. Finally, Step 8 defines the ``dframe`` function, which sorts and organizes the data into distinct datasets, including one that considers backorders—orders delayed due to inventory shortages or other issues—and one that does not. This allows for a comprehensive evaluation of the supply chain's efficiency and highlights areas where improvements are needed. The entire process creates a dynamic, transparent, and immutable system that automates the end-to-end order fulfillment

process, ensuring that all participants in the supply chain meet their obligations and that the supply chain operates smoothly and efficiently.

Sentiment Analysis Smart Contract Abbreviations

Variable	Abbreviation	Description
text	Text	The input text or review that is being processed.
categories	Categories	A dictionary containing descriptions of supply chain components.
preprocessed_categories	Preprocessed Categories	The dictionary with preprocessed text descriptions of the categories.
distilbert_model_name	DistilBERT Model Name	The name of the DistilBERT model used for sentiment analysis.
sentiment_tokenizer	Sentiment Tokenizer	The tokeniser is used with the DistilBERT model for sentiment analysis.
sentiment_model	Sentiment Model	The DistilBERT model is used for sentiment analysis.
sentiment_pipeline	Sentiment Pipeline	The pipeline for processing sentiment analysis using DistilBERT.
bert_model_name	BERT Model Name	The name of the BERT model used for category classification.
bert_tokenizer	BERT Tokenizer	The tokenizer is used with the BERT model for category classification.
bert_model	BERT Model	BERT is used for generating embeddings for category classification.
inputs	Inputs	Tokenized input text passed to the BERT model.
outputs	Outputs	The output embeddings are generated by the BERT model.
embeddings	Embeddings	The token embeddings mean from the last hidden layer of the BERT model.
review	Review	A single review or sentence is being analyzed.
preprocessed_review	Preprocessed Review	The review after it has been processed and cleaned.
sentiment_result	Sentiment Result	The result of sentiment analysis on the review.
sentiment_label	Sentiment Label	The sentiment label (e.g., positive, negative) from the sentiment analysis.
category_embeddings	Category Embeddings	The embeddings for the category descriptions were generated by BERT.
review_embedding	Review Embedding	The embedding of the preprocessed review generated by BERT.
cosine_similarities	Cosine Similarities	The cosine similarity scores between the review and category embeddings.
best_category_index	Best Category Index	The index of the category with the highest cosine similarity.
best_category	Best Category	The category corresponds to the highest cosine similarity score.
file_path	File Path	The path to the Excel file containing review data.
df	DataFrame	The DataFrame loaded from the Excel file containing review data.
ful_rev	Full Review	The full review text from the DataFrame.
reviews	Reviews	The individual sentences were extracted from the full review.
rows	Rows	A list of dictionaries where each dictionary represents a processed review.
data	Data	The final DataFrame contains processed reviews, sentiments, and categories.
X_train	X Train	The training set of reviews.
X_test	X Test	The testing set of reviews.
y_train	Y Train	The training set of sentiments and categories.
y_test	Y Test	The testing set of sentiments and categories.
sentiment_encoder	Sentiment Encoder	The label encoder is used to encode sentiment labels.
category_encoder	Category Encoder	The label encoder is used to encode category labels.

augmenter	Augmenter	Text augmenter is used to generate additional samples for balancing classes.
sentiment_category_counts	Sentiment-Category Counts	The counts of samples in each sentiment-category pair.
target_count	Target Count	The samples maximum count in sentiment-category as a balancing target.
augmented_texts	Augmented Texts	The list of original and augmented texts used to balance the classes.
augmented_sentiments	Augmented Sentiments	The list of sentiments corresponds to the augmented texts.
augmented_categories	Augmented Categories	The list of categories corresponds to the augmented texts.
current_count	Current Count	The number of samples in the current sentiment-category pair.
deficit	Deficit	The difference between the target and the current count for balancing.
original_text	Original Text	The original review text before augmentation.
augmented_text	Augmented Text	The augmented text is generated by the augmenter.
augmented_df	Augmented DataFrame	The final DataFrame contains the augmented and balanced data.
BERT	Bidirectional Encoder Representations from Transformers	Pre-trained model used for text classification tasks.
Embedding	Embedding Layer	An Embedding layer in neural networks converts text into dense, fixed-size vectors. These vectors represent words in a lower-dimensional space and capture the text's semantic meaning, serving as input to models like BERT.
API	Application Programming Interface	A set of rules that allows different software entities to communicate. Used here to fetch Amazon reviews.
GPU	Graphics Processing Unit	A hardware unit optimized for parallel processing, used for speeding up computations in deep learning tasks.
CUDA	Compute Unified Device Architecture	A parallel computing platform and programming model by NVIDIA that allows using GPU for general-purpose processing.
API URL	Application Programming Interface Uniform Resource Locator	The specific address or endpoint to which API requests are sent.
JSON	JavaScript Object Notation	A lightweight data-interchange format used to parse and structure the API response.
pt	PyTorch Tensor	A data structure used in PyTorch to store multi-dimensional arrays for deep learning.
CPU	Central Processing Unit	The main processor of a computer, used here when a GPU is not available.
batch_size	Batch Size	The number of samples processed before the model's internal parameters are updated.
max_length	Maximum Length	The maximum number of tokens (words) allowed in the tokenized sequence.
logits	Logarithmic Odds	The raw predictions output by the model before applying an activation function like softmax.

SENTIMENT ANALYSIS SMART CONTRACT Algorithm

STEP 1: Define the function **PREPROCESS** with the parameter **TEXT**.

STEP 1.1: Convert **TEXT** to lowercase.

STEP 1.2: Remove URLs from **TEXT** that start with "http" or "https".

STEP 1.3: Replace multiple whitespace characters in **TEXT** with a single space.

STEP 1.4: Remove content within square brackets from **TEXT**.

STEP 1.5: Remove common pronouns (e.g., "it", "you", "they") from **TEXT**.

STEP 1.6: Remove newline characters from **TEXT**.

STEP 1.7: Remove words in **TEXT** containing digits.

STEP 1.8: Remove HTML tags from **TEXT**.

STEP 1.9: Replace contractions in **TEXT** (e.g., "ve", "m").

STEP 1.10: Remove punctuation from **TEXT**.

STEP 1.11: Remove non-ASCII characters from **TEXT**.

RETURN the cleaned **TEXT**.

STEP 2: Define the dictionary *CATEGORIES* with specific descriptions for each supply chain component.

STEP 2.1: Create a dictionary with keys "none", "retailer", "warehouse", and "supplier".

STEP 2.2: Assign detailed descriptions to each key in *CATEGORIES*.

STEP 2.3: Preprocess each description in *CATEGORIES* using the *PREPROCESS* function.

RETURN the preprocessed *CATEGORIES*.

STEP 3: Load the DistilBERT model for sentiment analysis.

STEP 3.1: Load the DistilBERT tokenizer and model fine-tuned on SST-2 for sentiment analysis.

STEP 3.2: Initialize a sentiment analysis pipeline using the loaded model and tokenizer.

STEP 4: Load the BERT model for category classification.

STEP 4.1: Load the BERT large cased tokenizer and model.

STEP 4.2: Define the function *GET_BERT_EMBEDDING* to obtain BERT embeddings for a given *TEXT*.

STEP 4.3: Inside *GET_BERT_EMBEDDING*, tokenize the input *TEXT* and generate embeddings using the BERT model.

STEP 4.4: Return the mean of the token embeddings from the last hidden layer.

STEP 5: Define the function *ANALYZE_REVIEW_DISTILBERT_BERT* to analyze sentiment and classify categories for a single review.

STEP 5.1: Preprocess the input *REVIEW* using the *PREPROCESS* function.

STEP 5.2: Perform sentiment analysis on the preprocessed *REVIEW* using the DistilBERT model.

STEP 5.3: Obtain BERT embeddings for the preprocessed *REVIEW* and the descriptions in *CATEGORIES*.

STEP 5.4: Calculate cosine similarities between the *REVIEW* embedding and the category embeddings.

STEP 5.5: Determine the best matching category based on the highest cosine similarity.

RETURN the preprocessed *REVIEW*, its sentiment label, and the best matching category.

STEP 6: Load and preprocess the review data from the Excel file.

STEP 6.1: Load the Excel file containing review data.

STEP 6.2: Drop unnecessary columns from the loaded DataFrame.

STEP 6.3: Split each full review into individual sentences.

STEP 6.4: For each sentence, analyze the sentiment and category using *ANALYZE_REVIEW_DISTILBERT_BERT*.

STEP 6.5: Store the results in a new DataFrame.

STEP 6.6: Filter out rows with the "none" category and reset the DataFrame index.

STEP 7: Define the function *AUGMENTANDBALANCE* to augment and balance the review data.

STEP 7.1: Split the data into training and testing sets.

STEP 7.2: Encode the sentiment and category labels using label encoders.

STEP 7.3: Initialize a synonym augementer using WordNet.

STEP 7.4: Calculate the maximum count of samples for any sentiment-category pair.

STEP 7.5: Augment the data to balance the classes by generating additional samples using the augementer.

STEP 7.6: Create a new DataFrame with the augmented and balanced data.

STEP 7.7: Decode the sentiment and category labels back to their original form.

RETURN the augmented and balanced DataFrame.

STEP 8: Evaluate and Report Results Using BERT Model

STEP 8.1: Predict the Sentiment and Category Labels Using the BERT Model

STEP 8.2: Evaluate the Model Performance

STEP 8.3: Generate and Visualise Confusion Matrices

STEP 8.4: Report Detailed Classification Metrics

STEP 8.5: Store Predictions for Analysis

STEP 8.6: Finalise and Save Results

STEP 9: Smart Contract for Fetching and Classifying Amazon Product

STEP 9.1: API Request and Response Handling

STEP 9.2: Sentence Splitting

STEP 9.3: Device Setup

STEP 9.4: Tokenisation and Classification

STEP 9.5: Result Compilation

STEP 9.6: Result Display

The first step in the sentiment analysis algorithm involves defining a function `PREPROCESS` that normalizes input text. It takes care of removing irrelevant elements such as URLs, punctuation, non-ASCII characters, and other irrelevant content, which will make the text consistent and ready for further processing. This is important preprocessing to get rid of noise in the data to prevent interference with the accuracy of the subsequent

analyses. Because of standardization, the algorithm cleans the text in such a way that the models will have the cleanest and uniform data fed into them, thus improving the reliability of the results.

In the step 2 defines a dictionary, `CATEGORIES`, to fill in specific descriptions for respective parts of the supply chain, such as "retailer," "warehouse," and "supplier." These categories represent the discrete parts of the supply chain that reviews may relate to. Associated descriptions for each category have also been preprocessed using the same `PREPROCESS` function to make them consistent with the text data that will be analyzed. This step is important because it ensures both the input text and category descriptions have been pre-processed in the same way for comparison and classification in later stages. Step 3: Load the pre-trained transformer model, DistilBERT, fine-tuned on Sentiment Analysis. DistilBERT is a leaner and faster variant of BERT, which retained much of the accuracy of BERT with a much lower computational cost. The model is loaded in this step along with its tokenizer to set up the sentiment analysis pipeline. The purpose of this step is to determine the class or sentiment of each text-either positive, negative, or neutral. This sentiment analysis helps in knowing the emotional tone of the customer's feedback, which is important to understand the context of reviews. Step 4: Loading the BERT model used for generating embeddings-numerical representations-of text. These embeddings capture the meaning of text semantically, and the algorithm can classify this text into one of the predefined supply chain categories. Whereas the main reason for BERT's popularity is understanding words in context-even out of context-the model works very effectively for tasks relating to text classification. The review text and the category descriptions are embedded, and the algorithm can then compare them using measures of similarity that would help in accurately assigning each review to the proper category.

For the step 5, the function `ANALYZE\_REVIEW\_DISTILBERT\_BERT` is defined: it combines both tasks-sentiment analysis and category classification of each review. This function first preprocesses the review by calling the `PREPROCESS` function and then feeds the review into the DistilBERT model to analyze the sentiment of the review. It does sentiment detection, just like before, but in this case, using the BERT model, the function categorizes the review into the appropriate supply chain category, based on the embedding similarity to the predefined categories. The dual analyses therefore provide a good insight into the emotional tone underlying each review and the specific supply chain component relating to each review. Step 6: Loading review data from an Excel file, preprocessing, which means that each review is split into individual sentences. Further, each sentence is checked using previously defined functions regarding sentiment and category. The result - sentiments and categories for each sentence - will be stored in the new DataFrame:. This structured data helps in the granular analysis of customer feedback to find specific issues related to different parts of the supply chain. This would be done through Step 7, `AUGMENTANDBALANCE`, to handle the class imbalance in the dataset. The function will employ augmentation techniques in the text to generate further samples for these sentiment-category pairs, which are less represented. This function augments the data in order to ensure the balance of a dataset that is very critical for training the machine learning models for high performance across all classes. Further, one can get more generalizable models out of a balanced dataset, which in turn helps in having more reliable insights into customer feedback within the supply chain context. It covers all facets of customer sentiment and feedback necessary for adding value in rethinking and re-strategising supply chain processes.

Moving on, Step 8 of the heuristic algorithm explains how to evaluate and report results from the BERT model. The results predicted as output for both sentiment and category labels with the help of the BERT model are further evaluated on different performance metrics such as accuracy, precision, recall, and F1-score. Confusion matrices are generated and visualized to depict model performance regarding the classification task of categories and sentiments. After that, detailed classification metrics were compiled and reported with the aim of getting an insight into this model's performance. Prediction output is stored, including metrics for analysis. This last step would ensure that the outcome is well documented and saved for future reference, hence giving a clear understanding of the model's effectiveness in processing and classification of the review data.

Finally, Step 9 presents a smart contract intended for fetching and classifying reviews of some Amazon products. The development of this step concerns the creation of an API request that would fetch the product reviews and effectively handle the response of such. The reviews fetched are split into sentences, each of which goes for processing in the BERT model for sentiment and category classification. The results from this classification are collated together in a structured format, such as a DataFrame, for display or saving. This step stitches the whole process into one workflow that fetches data through an API, goes through advanced machine learning models for classification, and then provides a robust solution for customer feedback analysis in a supply chain context. Each of these steps taken together under the heuristic algorithm allows for a comprehensive and automated approach to sentiment analysis and categorization of reviews so that businesses understand, to a very big degree of accuracy and efficiency, customer reviews.

4. Results Analysis and Discussion

4.1 Introduction

The next section will deliberate on and present the outcomes of experiments performed in light of sentiment and category classification within supply chain data. The analysis is important for comprehending how effective the methodologies and technologies are that are presently deployed to upgrade supply chain operations and sentiment analysis.

This chapter has two important findings sections: Blockchain and Sentiment. The section on Blockchain analyzes the effect of blockchain on the various constituents of the supply chain. It particularly looks into how infusion of blockchain affects the efficiency, transparency, and coordination of supply chains based on the time series analysis as well as comparative analyses of different levels of fulfillment and shortages across conditions that have blockchain in place versus those that do not. The results here show the huge benefit of blockchain to supply chain performance and the solution to some of the common problems.

This section is concerned with the performance of the various models, that is, LSTM and BERT, for the purpose of sentiment classification. It, therefore, goes ahead to provide a clear comparison in key metrics, which include accuracy, precision, recall, and the F1-score. These metrics remain very critical in assessing the models' ability to predict correctly the customer's sentiment in supply chain management. First, this section presents the results for each model; it goes on to give indications of the best model, hence guiding future applications in the domain of sentiment analysis.

4.2 Blockchain

The below time series graph shows the customer requests, retail offers, warehouse offers, and supplier offers relationship throughout 12-months period after blockchain being applied to the supply chain. The components showcased represent the critical supply chain part, and the observed trends offer useful information into the supply and demand dynamic.

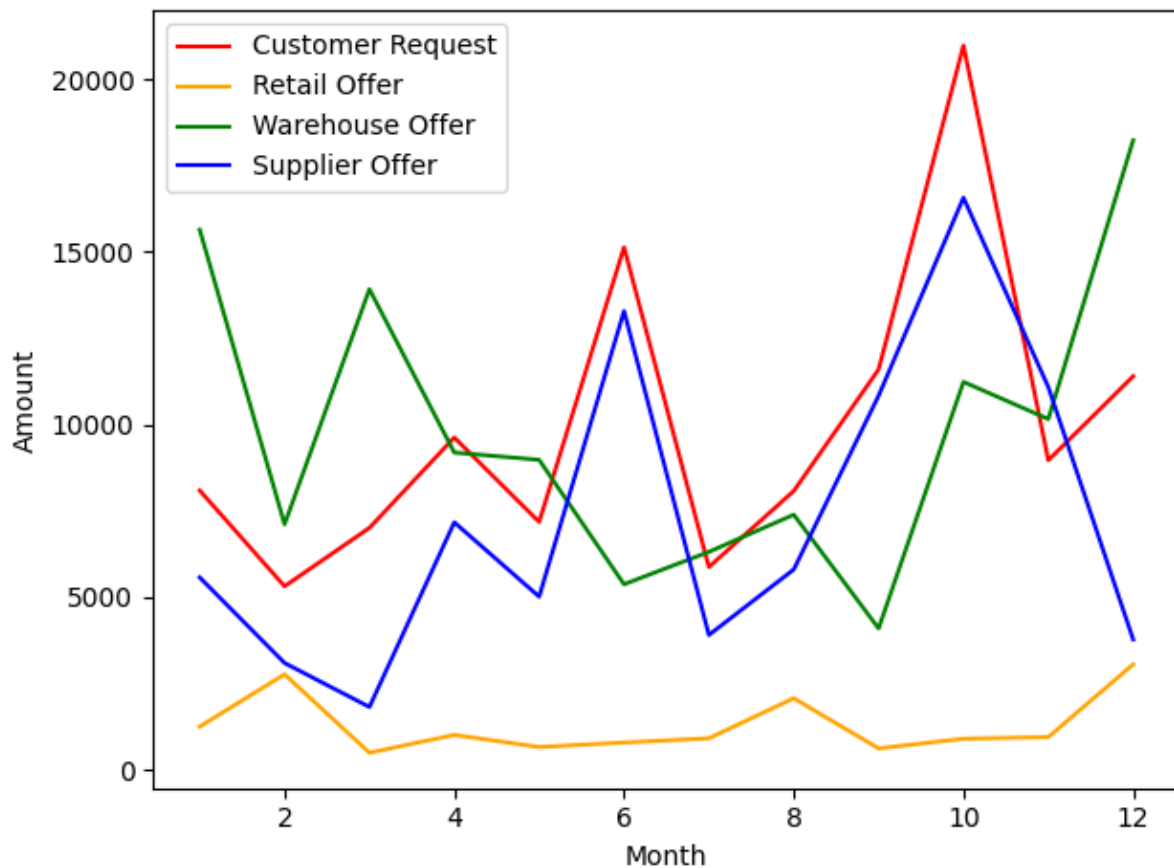


Figure. 7. Time series of customer demand to supply chain components' offering with blockchain

As shown by customer request (red) line, several peaks can be seen throughout the chart, showcasing the vast fluctuations and highly dynamic in customer demand over time, such as months 1, 6, 9, and 11, where month 11th shows the highest peak.

Next, the retail offers (orange) line stagnate at the lowest part of the chart, indicating its inability to fulfill most of the customer requests compared to any other component's offering. This happens due to the need of the retailers to keep their storage from overstocking and effectively manage their inventory costs, which resulted in a conservative yet steady approach to product stocking.

Then, a massive increase in demand fulfillment is shown by the warehouse (green) line in the first few months. Although the line is not consistently depicted that it is able to fulfill more demand in the upcoming months, but, the chart shows that warehouses can adapt and more responsive to the customer demand fluctuation despite its variability and lags. Its adaptability is shown through its peaks that occurred in months 2, 4, and 11, that somewhat aligns with the customer request peaks. All in all, warehouse responsiveness indicates its attempt to observed and followed demand trends and adjust their stock levels.

Last, the blue line, which represents the supplier offers, also presented variability, representing its reactivity to both customer requests and warehouse offers. Throughout the year, notable peaks can be seen in months 3, 6, and 9 in suppliers offer chart, which shows

more similarity to the customer demand more than warehouses. This phenomenon proves that the suppliers can adjust their outputs to the entirety of the supply chain signals rather than immediate customer demand alone.

As a whole, every component across the supply chain demonstrates their efforts to match the supply with the demand, with some noticeable mismatches and lags. Both supplier and warehouse offers provide a great attempt to meet customer demand, however, the magnitude and timing of these efforts may require more effort to align with the customer requests peaks. On the other hand, retail offers remain as the lowest component offer rate, indicating its more cautious approach to inventory management.

The supply chain shows to be highly dynamic, where each component is able to react to supply and demand changes, thanks to the help of blockchain. But, supply chain synchronisation delays and inefficiencies suggest that it can be improved. To better meet customer demand, every component in the supply chain can improve their responsiveness to be more aligned to it more efficiently and effectively, by intensively communicating to each other more, which is possible through the blockchain system.

To sum up, despite the great adjustability of each component with the help of blockchain, responsiveness and synchronisation improvement across the supply chain, supported by blockchain, are necessary to enhance overall efficiency and reduce mismatches.

The below pie charts compare the retailer, warehouse, supplier offers distribution and shortages in the supply chain between the one that used blockchain and the one that did not. The charts offer visualisations of how blockchain technology influences the resource allocation and supply chain efficiency.

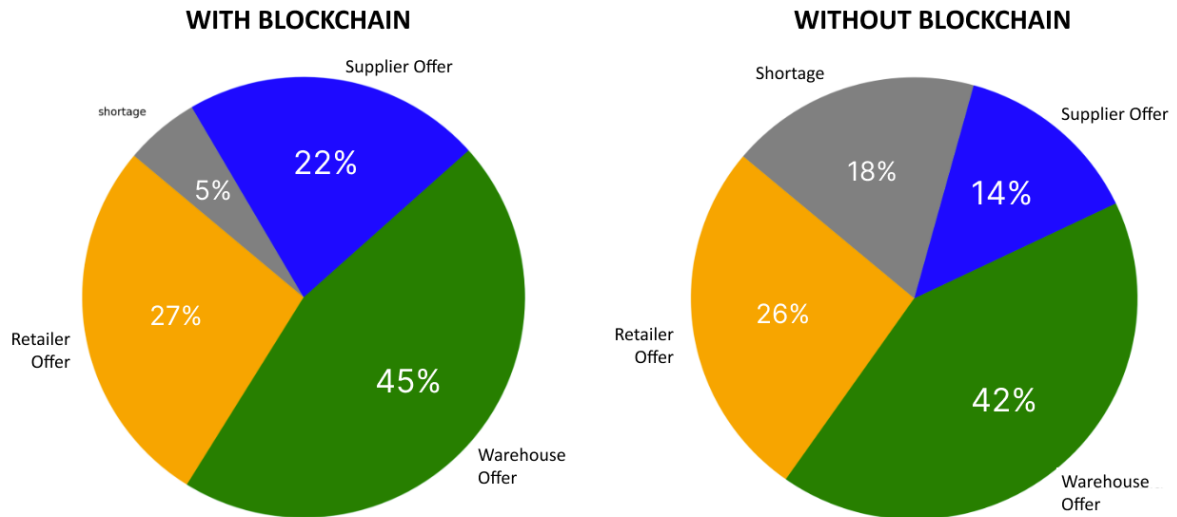


figure 8. Comparison of components fulfillment with and without blockchain

The chart titled “With Blockchain” on the left demonstrates the supply chain component distribution with blockchain integration. As seen in the chart, warehouse took the largest supply chain activity portion with a share of 45%. However, supplier and retailer offer share almost the same distribution of 22% and 27%. Notably, the shortage of offering can be pushed down to 5% with blockchain, proving its integration ability to help reducing inefficiencies and guaranteeing the balance across the supply chain.

On the other hand, the “Without Blockchain” chart on the right shows a different result to its counterpart. Warehouse still account for the biggest portion of 42% supply chain activities, however, the supplier fall much lower to 14% compared to the retailer which account for 26%. This happened due to the sudden rise in shortages that took up to 18%, 13% more compared to when blockchain technology was applied, indicating the inability of each component to coordinate efficiently.

The most important distinction of the two scenarios is the difference in shortages in supplier fulfillment share. Thanks to blockchain integration, it helped significantly decrease the shortage from 18% when the technology is absent, to 5%. This is made possible due to the ability of blockchain technology to offer an enhancement in coordination and transparency that allows reduction in shortages occurrences and better allocation of resources.

Nonetheless, the other components of the chart had a slight to huge increase in activities. For example, supplier contributed more to the supply chain, as proven with the increase of distribution from 14% to 22% due to better integration and responsiveness to demand signals. Afterwards, slight increase in activity happened to the retailer offers from 26% to 27% that suggest an efficient and more balanced distribution system. Also, warehouse offers maintain its large portion of contribution, with a small increase from 42% to 45% when integrated with blockchain, suggesting that warehouses can still improve their efficiency and manage inventory better when being supported by blockchain technology, despite the modest rise.

Blockchain integration had significantly boosted supply chain efficiency through the reduction of shortages and increase in suppliers and retailers responsiveness as shown by the

two scenarios comparison. A more balanced and efficient distribution was made possible by the improved transparency and coordination of resources distribution in the entire supply chain environment. The blockchain technology potential to optimise overall performance and address key supply chain challenges are evident in the visual representation.

Subsequently, another charts that show each component monthly fulfillment of the customer requests over a 12-month period, comparing with and without blockchain implementation. Both of the charts offer a particular view of different supply chain components contributed to cater customer demand and the blockchain technology impact on reducing shortages.

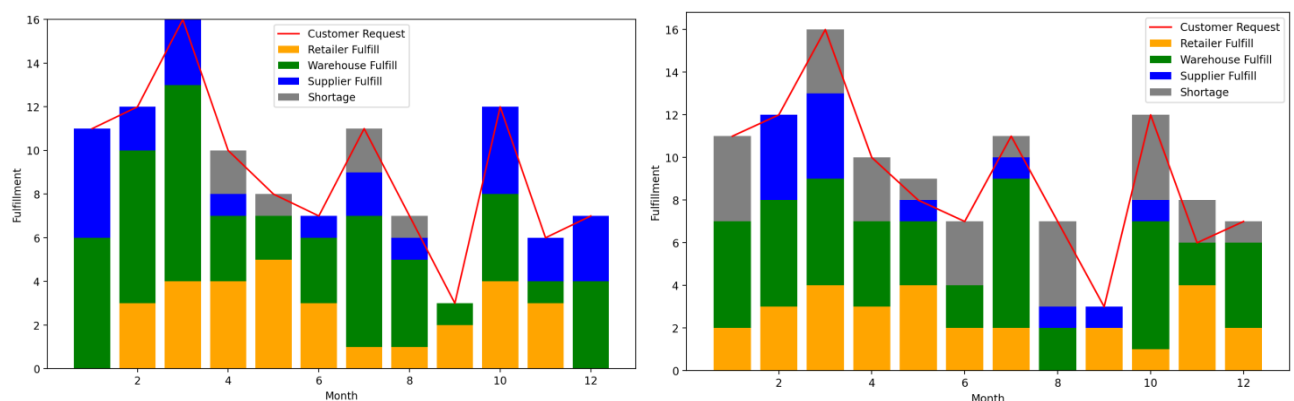


Figure 9. Comparison of Monthly Fulfillment and Shortages in Supply Chain with and without Blockchain Integration

On the left, the illustration demonstrate how each of the component fulfilled customer request after blockchain is implemented. Each of the components, such as customer demand (red line), retailers (orange bar), warehouses (green bar), suppliers (blue bar) and the resulting shortages (gray bar) are displayed. As evident in the “With Blockchain” chart (left), synchronisation is more prevalence between the supply chain components that lead to smaller amount of shortages and better customer demand alignment. Especially, in the 2nd, 3rd, and 10th month shows significant contributions from all components, resulting in minimal shortages.

Meanwhile, the bar chart on the right shows how each supply chain components fulfill customer requests without blockchain technology. In this case, shortages are more noticeable across the year, especially in the 6th, 8th, and 10th month, showcasing the inefficiency in coordination between supply chain components and higher unmet customer demand frequencies.

As evident, the critical difference between the two cases is the shortages level, where vast discrepancies are clearly apparent between the one with and without blockchain. With the help of blockchain, the gray bars (shortages) can be reduced significantly. This huge reduction proposes that blockchain implementation improves coordination and transparency in the supply chain that allows for more effective customer requests fulfillment.

Higher values and better stability is shown by the retailer and warehouse in the blockchain-integrated scenario which showcased that the mentioned component could manage their inventory and respond to demand signals better. Also, an increase in supplier contribution was shown in the blockchain-integrated chart that better balanced the supply chain.

At the same time, larger and more frequent shortages occurred when blockchain is not integrated to the supply chain due to the lack of timely information sharing and inefficiencies, resulting in higher unmet customer demand level due to the unresolved challenges typically faced by supply chain components in the absence of advanced coordination technologies.

The two scenarios comparison illustrates that significant improvement in supply chain efficiency is made possible by blockchain implementation due to the advance suppliers, warehouses, and retailers coordination and shortages reduction. The visualisation underscores blockchain's importance to enable supply chain performance optimisation to make sure that each supply chain component can better align with customer demand and lessen shortages frequency and magnitude that leads to better supply chain efficiency and responsiveness that allows them to consistently achieve customer expectations.

Large Language Model Sentiment Analysis

Data Preprocessing and Exploration

The following figure better illustrates the distribution of sentiment based on the three major bases of the supply chain: retailer, warehouse, and supplier. In this light, from the data one would find that there is indeed a serious imbalance to each sentiment and category. For example, the positive sentiments becomes completely dominated by the retailer section with 503 instances and commands 33.76% of the dataset. This domination would indicate that most of the customers' feedback is spent in positively speaking about experiences they have had with the retailers. This may be due to a few factors that it may involve, such as customer service, product availability, or general shopping experience. Comparatively, both the warehouse and supplier are also highly rated categories but very much lower. It means it could be because this form of interaction does not happen as often, or the nature of it tends to less impactful when it relates to a positive review. This is demonstrated by the lower status of the two that shows a far lower level of direct interaction with end consumers.

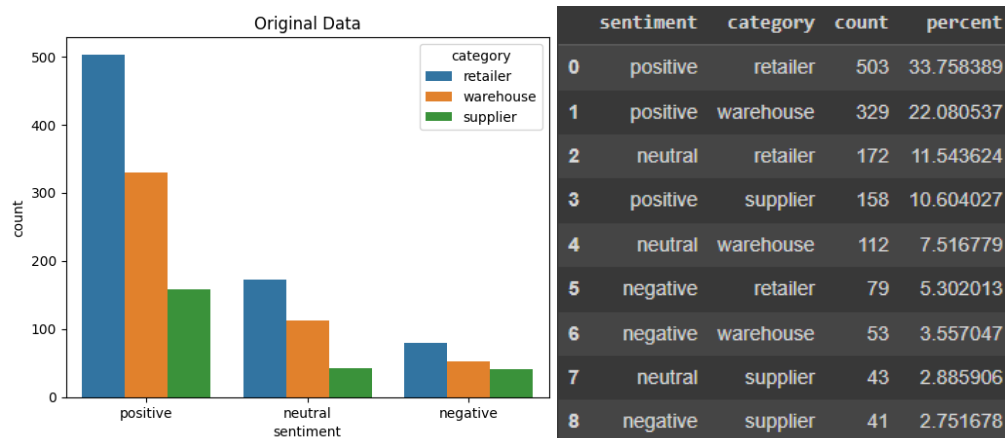


Figure 10. Original Data Sentiment and Component Distribution

The other axis, with neutral and negative categories of sentiments, is slightly underrepresented by all three components. Again, the neutral sentiment is more balanced than the negative, outsized, and in this case, especially in suppliers, perhaps indicating no strong feeling toward these suppliers. This could be a function of the fact that all of the interactions customers have with the suppliers are not quite as directly visible and influential as compared to the end consumer with respect to retailers.

The negative sentiment category is very thin, mainly with very low counts in both the warehouse and supplier segments, probably due to fewer problems reported, or the negative experiences get mitigated before they even get to the consumers. All these reflect a kind of imbalance across sentiment categories and supply chain components, the latter turns into another critical challenge in the original dataset that may bias any machine learning model which might be trained by this data.

There could be some overweighting of positive retailer experiences and hence bias the view toward seeing an inaccurate prediction or underestimation of neutral or negative feedback. It contains such skewness that, unless dealt with, can be particularly harmful to the model's ability to provide balanced views and rich insights into, among other things, developmental points of a supply chain. Hence, mitigating this imbalance becomes very important in order to make predictions of the model worthy and as an instance for the entire spectrum of customer sentiment.

Imbalance in the dataset was corrected using Data Augmentation and Balancing. The result after the application of Data Augmentation and Balancing on the original imbalanced dataset is a very important factor while fine-tuning the accuracy and reliability of the sentiment analysis model. This is because it creates synthetic examples for the underrepresented classes, thereby artificially increasing their presence within the data set. While oversampling by random means involves multiple copying of available instances, Data Augmentation and Balancing creates new plausible samples through the interpolation of existing examples. That is, an enrichment of a dataset without replicating the same points over and over again.

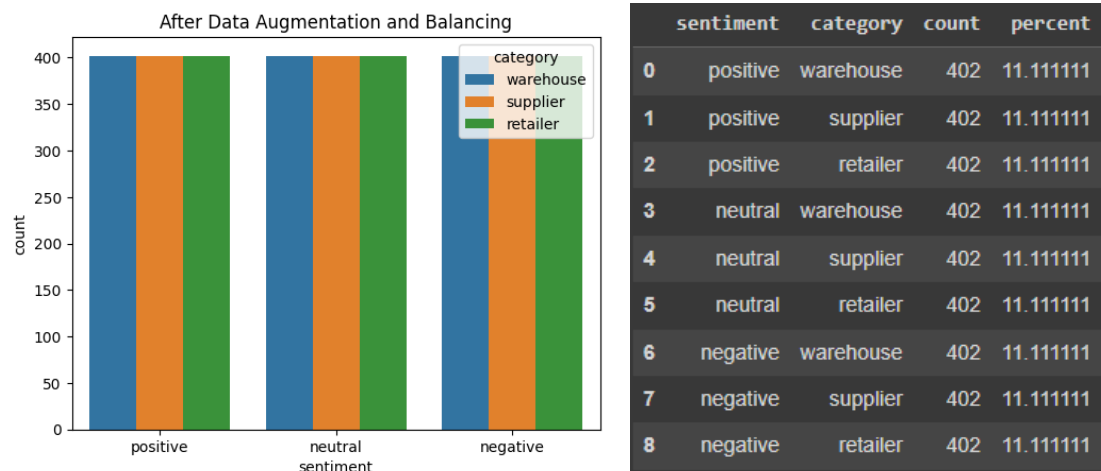


Figure 11. Augmented Data Sentiment and Component Distribution

Therefore, the function can modulate the already skewed sentiment distribution concerning the retailer, wholesaler, and supplier categories. Now, each class will have exactly 402 instances for all three supply chain components. Balancing like this rectifies the exit and possible over-representations of positive sentiment in the original dataset. Furthermore, it will ensure that equal learning from various sentiments occurs that will reduce most bias risks toward those more common classes for sentiment.

This balancing of the dataset using Data Augmentation and Balancing will increase the model's accuracy and sensitivity to customer sentiment classification and help on the blockchain scalability to distribute the knowledge to different supply chain components. For example, negative feedback regarding suppliers, which was underrepresented in an imbalanced dataset, is now well represented, and hence the model will learn and predict these sentiments as accurately as it does with those positive or neutral. This extensive training will allow such models to have more unaws in view of actionable insights and, hence, leading to more informed decision-making processes in the management of supply chain activities.

This visualization is a qualitative counterpart to the above-discussed quantitative sentiment analysis. If the sentiment analysis gives structured insight into customer opinions with respect to being positive, neutral, or negative, the word cloud expresses a more freeform view of specific topics and keywords dominating customer discourses. These tools jointly provide an overview of the customer feedback and thus help supply chain managers and business corporations make informed decisions toward better serving the needs and expectations of customers.

The below tables illustrate how sentiment analysis and component categorization work in customer reviews against the background of a supply chain. The tables provide different combinations of sentiment, positive, neutral, and negative, and supply chain components, retailer, warehouse, and supplier, explaining how a full review will be processed and broken down into more focused and related reviews.

Sentiments	Positive
Component	Retailer
Full Review	Great Govee quality product. I have other products from this company, and they all worked great. The installation was medium to easy, depending on the level of experience you have. Because I have other products from them, adding it to the app was easy. It has controls on it also so someone without the APP can still control it. It integrates great with Amazon Alexa and Google Home. It is very versatile and the colors are great. With Govee, you pay a little more than other companies, but you get great quality products. I have other, cheaper ones, but the quality difference is big. We worth the extra money for extra piece of mind.
Related Review	We worth the extra money for extra piece of mind
Focused Review	worth the extra money for extra piece of mind

Sentiments	Negative
Component	Supplier
Full Review	I love the lights. Being able to control the color is a lot of fun and even as normal lights the different tones between warm and cool white is really pleasant. My only complaint is the app took an hour to even make it fully function. It needed a firmware update but refused to apply it until I exited out and went into the top right drop down to do it. The main prompt to update did not work at all. It also keeps asking for location tracking which shouldn't be needed for a lamp app.
Related Review	My only complaint is the app took an hour to even make it fully function
Focused Review	my only complaint is the app took an hour to even make fully function

Sentiments	Neutral
Component	Warehouse
Full Review	I love the lights. Being able to control the color is a lot of fun and even as normal lights the different tones between warm and cool white is really pleasant. My only complaint is the app took an hour to even make it fully function. It needed a firmware update but refused to apply it until I exited out and went into the top right drop down to do it. The main prompt to update did not work at all. It also keeps asking for location tracking.
Related Review	It also keeps asking for location tracking
Focused Review	also keeps asking for location tracking

In the first table, the sentiment is classified as Positive, and the component connected with the review is the Retailer. This whole review gives an example of the quality of one of Govee's products and praises the ease of use, integration with other devices, and the value perceived for the money paid. The related and focused reviews extract the essence of customer satisfaction, stressing that this product is well worth any extra money for the peace of mind alone.

The second example table portrays a review with a Neutral sentiment related to the Warehouse component. The full review is pretty satisfied with the features of the product but cites one large problem with the app, mainly in regard to updating and location tracking, which has become too frequent. These reviews relate to and focus only on the root of the customer's concern: the app that keeps asking to track the user's location, some element of the product that detracts from the user's experience.

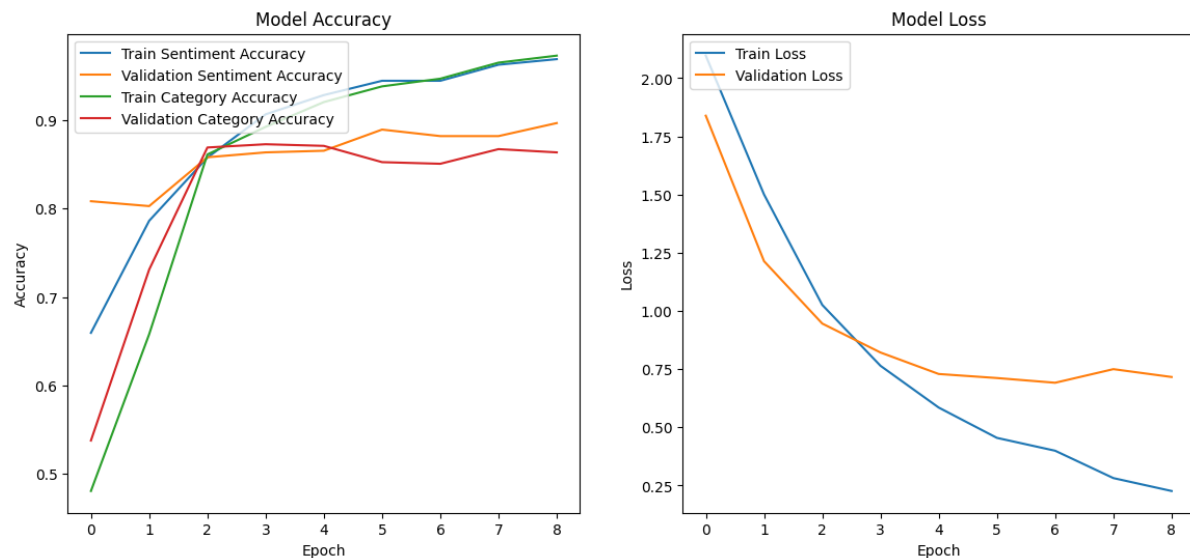
In the third table, the sentiment is Negative, and the review belongs to the Supplier component. Customer dissatisfaction is revealed in the review because of numerous problems in this application, mainly about the time it took for the app to start working well and demanding access to location tracking, which is useless. Related and focused reviews concentrated the complaint, clear about what had gone wrong: a too-long time taken to make the app fully functional.

These examples demonstrate how sentiment analysis, together with component categorisation, allows filtering to the most relevant and impactful constituent parts of customer feedback. This will not only identify the general sentiment of a review but also link it to a certain part of the supply chain, therefore able to enable targeted analysis and response.

Models Development

Long Short-Term Memory (LSTM)

The graphs above present the performance of an LSTM model during the eight epochs in terms of accuracy and loss for the training phase and the validation phase.



In the model accuracy graph, the blue line tracks the training sentiment accuracy, which lies around 0.55 at the beginning and shows a rapid improvement, then stabilizes above 0.9 by the third epoch, thus evidencing how the model quickly learns to classify sentiments effectively during training. The orange line of validation sentiment accuracy follows very closely, thus proving generalization on unseen data, until it finally attains a stable accuracy above 0.8 after a few epochs.

The green line in the same graph indicates training accuracy for category classification, which increases steadily, stabilizes above 0.9, and shows how well the model has learnt to perform this task. In contrast, the red line, which represents the validation category accuracy, becomes too variable. Although it did increase initially, it afterwards started oscillating around 0.8 after a few epochs. This could have meant that the model was actually having a harder time with category classification, or that the model was starting to overfit the training data.

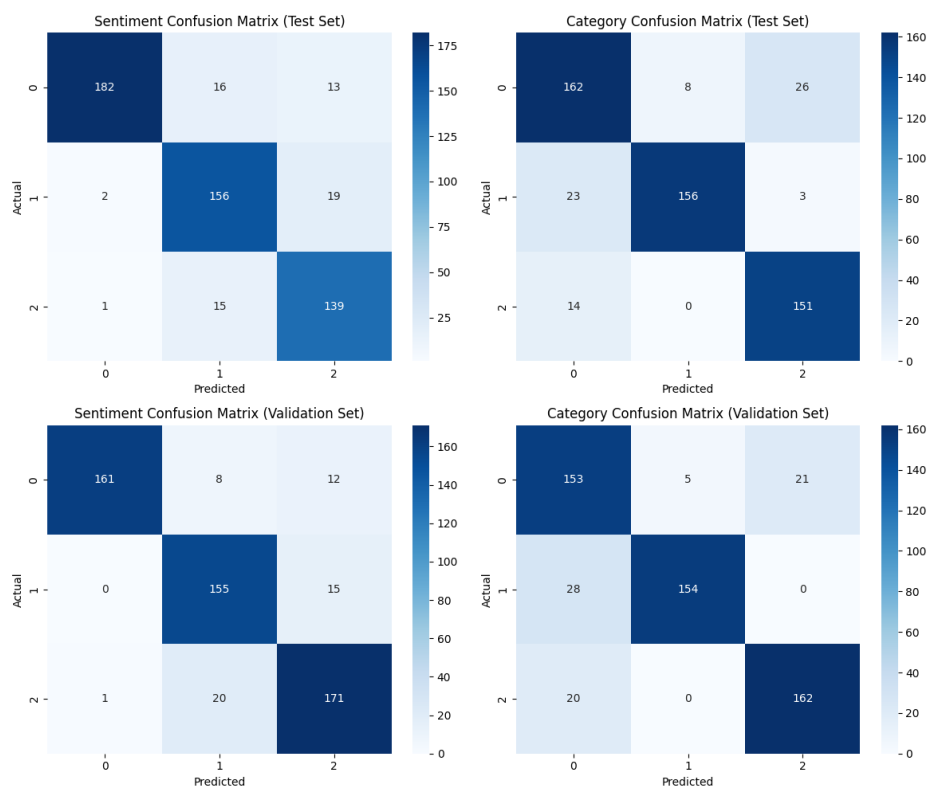
On the right is the model loss graph, showing the performance of the model. In blue, the training loss decreases continuously as the model learns; that is, it becomes increasingly accurate in its predictions with time. The training loss decreases to a very low level by the eighth epoch, correlating to the high accuracy observed in the accuracy graph.

The validation loss, the orange line, is decreasing first, which is good— it means the model performs well on unseen data. Then, it reaches a plateau around the third epoch and increases slightly. This behavior of the training and validation losses is the typical behavior in the cases of overfitting.

The LSTM model performed really well during training, with high accuracy and low loss for both sentiment and category classification. However, the results obtained on validation

showed that this is a potential case of overfitting after a few epochs in category classification. Overfitting is a critical observation since it would impact the real-world application performance, and therefore regularization or early stopping strategy might be needed for better generalisation.

The confusion matrices shown for the LSTM model provide the necessary information regarding the model performance in the prediction of sentiment and category on the test set and validation set.



Moving on to the sentiment confusion matrices, at the top left will be the test set and at the bottom left, the validation set. Each matrix contains the model's ability to predict correctly on the sentiment of three classes: negative as 0, neutral as 1, and positive as 2. It shows that, in general, the model did quite well on the test set—most are on the diagonal, meaning the actual sentiment matched the predicted. Especially, it correctly predicted 182 of the 211 negative sentiments, 156 of the 177 neutral sentiments, and 139 of the 155 positive sentiments. However, there are some misclassifications, mostly between neutral and positive sentiments, indicating that the model often gets confused between these two classes. Similar trends are witnessed in the validation set, where it has still been able to predict most of the sentiments correctly but still has some confusion, mostly between the neutral and positive classes.

On the right, category confusion matrices are shown. The top-right matrix is for the test set, and the bottom right one is for the validation set. These matrices depict how well the model separates according to the three categories: warehouse (0), retailer (1), and supplier (2). The model performs well on the test set, particularly while predicting the retailer

category. It has rightly predicted 156 out of 182 cases. Although there is some amount of confusion between the warehouse and supplier categories, this is seen when 23 instances are wrongly predicted as retailer while they actually belonged to the warehouse category and 14 as supplier while they were actually of the warehouse category. This trend is also replicated in the validation set, where in general, the model does well, but there is still some struggle in differentiating between certain categories, mostly between warehouse and supplier.

These confusion matrices reflect that although the LSTM model is generally effective in both sentiment and category classification, surely there is a scope of improvement in these models, particularly regarding the reduction of confusion between similar classes, such as neutral and positive sentiments or warehouse and supplier categories. This can be improved upon by further fine-tuning or including more data to enable the model to make better differentiation between these overlapping cases.

The classification reports for the LSTM model provide a detailed breakdown of the model's performance in terms of precision, recall, and F1-score for both sentiment and category classification on the test and validation sets.

Classification Report for Sentiment (Test Set):					Classification Report for Sentiment (Validation Set):				
	precision	recall	f1-score	support		precision	recall	f1-score	support
0	0.98	0.86	0.92	211	0	0.99	0.89	0.94	181
1	0.83	0.88	0.86	177	1	0.85	0.91	0.88	170
2	0.81	0.90	0.85	155	2	0.86	0.89	0.88	192
accuracy			0.88	543	accuracy			0.90	543
macro avg	0.88	0.88	0.88	543	macro avg	0.90	0.90	0.90	543
weighted avg	0.89	0.88	0.88	543	weighted avg	0.90	0.90	0.90	543

Classification Report for Category (Test Set):					Classification Report for Category (Validation Set):				
	precision	recall	f1-score	support		precision	recall	f1-score	support
0	0.81	0.83	0.82	196	0	0.76	0.85	0.81	179
1	0.95	0.86	0.90	182	1	0.97	0.85	0.90	182
2	0.84	0.92	0.88	165	2	0.89	0.89	0.89	182
accuracy			0.86	543	accuracy			0.86	543
macro avg	0.87	0.87	0.87	543	macro avg	0.87	0.86	0.87	543
weighted avg	0.87	0.86	0.86	543	weighted avg	0.87	0.86	0.87	543

Starting with the sentiment classification on the test set (left side), the model demonstrates strong performance across the three sentiment classes: negative (0), neutral (1), and positive (2). The precision, which measures how many of the predicted positive cases are actually positive, is notably high for the negative sentiment at 0.98. However, it is slightly lower for neutral and positive sentiments at 0.83 and 0.81, respectively. The recall, which indicates how many actual positive cases were correctly identified by the model, shows a similar trend with a high recall for negative sentiment at 0.86, and slightly lower recall for neutral and positive sentiments at 0.88 and 0.90. The F1-score, which is the harmonic mean of precision and recall, follows this pattern, with the model achieving a higher score for negative sentiment (0.92) compared to neutral and positive sentiments (0.86 and 0.85, respectively). The overall accuracy for sentiment classification on the test set is 0.88.

For the category classification on the test set, the model also performs well across the three categories: warehouse (0), retailer (1), and supplier (2). Precision is particularly high for the retailer category at 0.95, while it is slightly lower for warehouse and supplier at 0.81 and 0.84, respectively. Recall follows a similar trend, with the model recalling 0.83 for warehouse, 0.86 for retailer, and 0.92

for supplier. The F1-scores are consistent with these metrics, reflecting the model's strong performance in predicting the retailer and supplier categories, while it performs moderately well in predicting the warehouse category. The overall accuracy for category classification on the test set is 0.86.

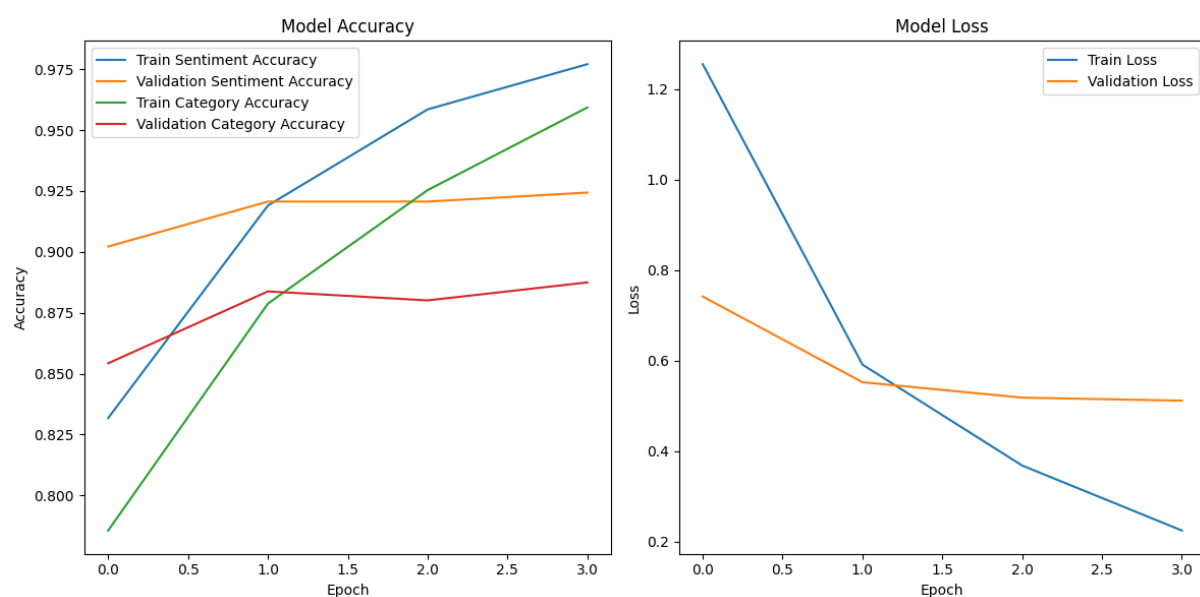
On the validation set (right side), the sentiment classification performance shows improvements across all metrics compared to the test set. The precision for the negative sentiment reaches 0.99, while it is slightly lower for neutral and positive sentiments at 0.85 and 0.86, respectively. The recall is also high across all sentiments, particularly for the positive sentiment at 0.89. The F1-scores mirror these results, with the model achieving an overall accuracy of 0.90 on the validation set, indicating that it generalizes well.

For the category classification on the validation set, the model maintains consistent performance with an overall accuracy of 0.86, similar to the test set. Precision, recall, and F1-scores are comparable to those on the test set, indicating that the model's ability to classify categories remains stable across different data splits.

In summary, the LSTM model demonstrates robust performance in both sentiment and category classification tasks, with particularly strong results in the retailer category and negative sentiment classification. The model performs consistently across both the test and validation sets, with only slight variations in metrics, suggesting that it generalizes well to new data.

Bidirectional Encoder Representations from Transformers (BERT)

The following two graphs represent the performance of a BERT model on training and validation, based on accuracy and loss, for several epochs. To the left are the accuracy diagrams of the model taken by the training epochs for sentiment and category classification, and to the right are the corresponding loss diagrams of the model.



The graph shown in the accuracy plot (left), it evidently changes to increased performance with each epoch of the model in the training and validation sets. The sentiment accuracy for training is in blue, starting pretty high and going up steadily to almost 0.975. Thus, by epoch

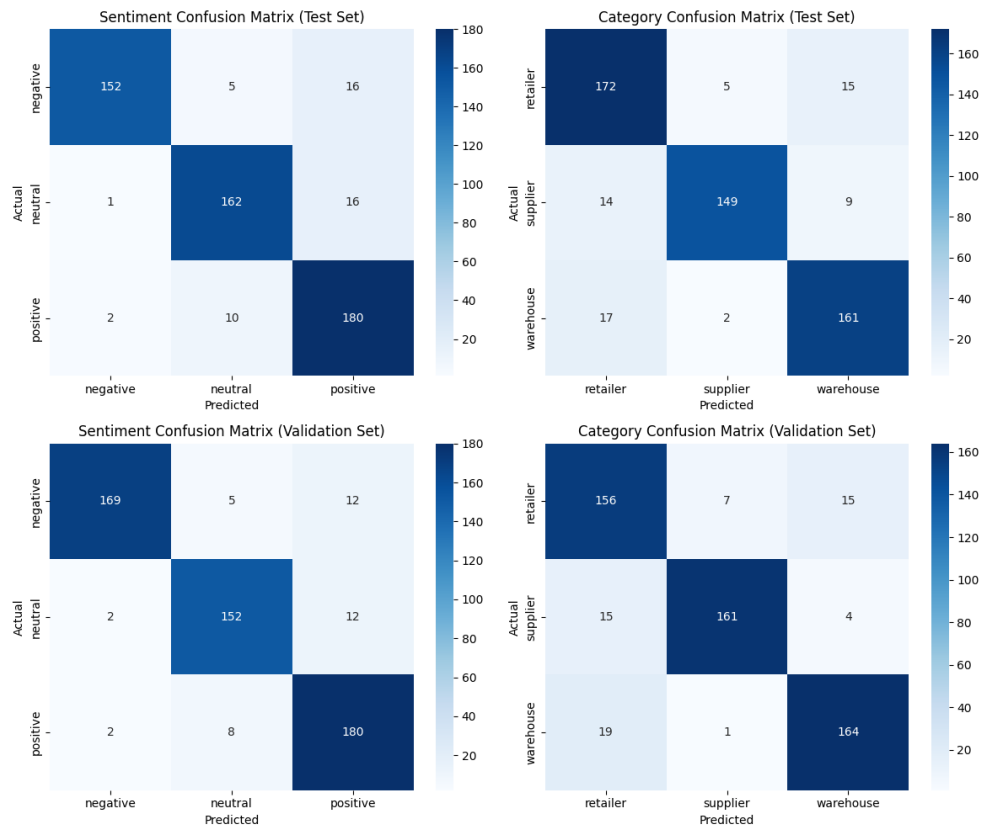
3, it clearly appears that the model is really learning lessons from training. The orange line of the validation sentiment accuracy peaks in a similar upward trend to around 0.95 by the third epoch. To be more precise, there is a good alignment between training and validation accuracy, which suggests that the model generalizes well to unseen data without overfitting.

It also tracks category prediction, where training category accuracy (green line plot) is slightly below sentiment but continues to increase. The validation category accuracy also gradually increases and remains below the training accuracy. By the third, the validation category accuracy has climbed above 0.90, reflecting good model performance but not as good as the sentiment classification.

In the right-hand graph, more information about the model training process is given. The training loss is the blue-lined curve, falling off in the course of epochs. The fact proves obviously that the model is minimizing the error on the training set. At the same time, the validation loss is also sharply decreasing, in comparison to the training loss. After the third epoch, the value of the validation loss starts to flatten slightly, indicating that the model is still learning something, but the improvements in reducing error on the validation set become close to negligible.

Overall, the BERT model performs really well on both sentiment and category classification tasks, since there is a rise in accuracy and a fall in loss in the epochs of training. The close alignment of the training validated metrics clearly shows this model generalizes well, avoids overfitting significantly, and can handle new data robustly.

The given confusion matrices contain all the information about the performance of the BERT model in both sentiment and category classification tasks across the test and validation datasets.



First, the sentiment classification's confusion matrix, one can note that most of the predictions are concentrated on the diagonal, showing correct classification. In particular, it correctly predicted 152 negative, 162 neutral, and 180 positive sentiments among the total predictions. However, some misclassifications occurred on the line of neutral and positive sentiments, whereby some of the positive sentiments were misclassified as neutral and vice versa.

The sentiment confusion matrix on the validation set charts a very similar trend. The model still performs very well, correctly predicting 169 instances of negative sentiment, 152 of neutral sentiment, and 180 of positive sentiment. It misclassified at almost the same rates compared to the test-set results, thus it showed consistency in performance across different data sets.

For category classification, the confusion matrix for the test set reveals that the BERT model is highly accurate in predicting the correct supply chain component, whether retailer, supplier, or warehouse. Most of the predictions are correct: 172 categories for retailers, 149 for suppliers, and 161 for warehouses are accurately classified. The largest mistakes are between the retailer and warehouse categories, some examples being misclassified as one while belonging to the other.

The category confusion matrix for the validation set reveals similarly, strong performance for this model to correctly identify 156 retailer, 161 supplier, and 164 warehouse categories. Although some misclassifications still occur, mostly between retailer and warehouse categories, the overall accuracy is very high.

These confusion matrices only go to show the excellent performance of the BERT model in both sentiment and category classification. In most cases, it performs with a very high degree of accuracy, in that most of the predictions are rightly classified, with very few errors, thus quite effective in handling complex classification tasks across different datasets.

Classification Report for Sentiment (Test Set):					Classification Report for Sentiment (Validation Set):				
	precision	recall	f1-score	support		precision	recall	f1-score	support
negative	0.98	0.88	0.93	173	negative	0.98	0.91	0.94	186
neutral	0.92	0.91	0.91	179	neutral	0.92	0.92	0.92	166
positive	0.85	0.94	0.89	192	positive	0.88	0.95	0.91	190
accuracy			0.91	544	accuracy			0.92	542
macro avg	0.91	0.91	0.91	544	macro avg	0.93	0.92	0.92	542
weighted avg	0.91	0.91	0.91	544	weighted avg	0.93	0.92	0.92	542

Classification Report for Category (Test Set):					Classification Report for Category (Validation Set):				
	precision	recall	f1-score	support		precision	recall	f1-score	support
retailer	0.85	0.90	0.87	192	retailer	0.82	0.88	0.85	178
supplier	0.96	0.87	0.91	172	supplier	0.95	0.89	0.92	180
warehouse	0.87	0.89	0.88	180	warehouse	0.90	0.89	0.89	184
accuracy			0.89	544	accuracy			0.89	542
macro avg	0.89	0.89	0.89	544	macro avg	0.89	0.89	0.89	542
weighted avg	0.89	0.89	0.89	544	weighted avg	0.89	0.89	0.89	542

The classification reports provided for the BERT model illustrate its performance in predicting sentiments and categories on both the test and validation datasets.

Starting with sentiment analysis on the test set, the model exhibits strong performance across all three sentiment classes—negative, neutral, and positive. The precision, recall, and F1-scores are particularly high, with precision ranging from 0.85 for positive sentiments to 0.98 for negative sentiments. The overall accuracy for sentiment prediction on the test set is 0.91, indicating that the model correctly predicts sentiment in 91% of the cases. The macro and weighted averages for precision, recall, and F1-score are all around 0.91, further confirming the model's reliability.

On the validation set for sentiment, the BERT model continues to perform well, even slightly improving with an accuracy of 0.92. Precision and recall values remain high, with the F1-score for positive sentiments improving to 0.92. This consistency across the test and validation sets suggests that the model is generalizing well to unseen data, maintaining its accuracy and precision across different datasets.

Moving on to category classification, the model also demonstrates robust performance. On the test set, the model achieves an accuracy of 0.89, with precision, recall, and F1-scores being strong across the retailer, supplier, and warehouse categories. The F1-scores are particularly high for the supplier category at 0.91, indicating that the model is highly effective at predicting this class. The macro and weighted averages for category classification are consistent, both sitting at 0.89.

The validation set for category classification shows similar results, with an overall accuracy of 0.89. The model maintains strong precision, recall, and F1-scores across all categories, with slight variations but no significant drop in performance. The consistency in these metrics across both the test and validation sets again emphasizes the model's ability to generalize well and accurately predict categories in diverse contexts.

Overall, these classification reports underscore the BERT model’s effectiveness in handling both sentiment and category classification tasks with high accuracy and reliability across different datasets. The model's strong performance metrics suggest it is well-suited for these types of classification challenges, providing precise and consistent results.

Comparison Study

The tables below representing the performance of various models: BERT, LSTM, SVM, Random Forest, and KNN. These are evaluated for two tasks: Sentiment Classification and Component Classification.

SENTIMENT CLASSIFICATION MATRICES				
MODEL	ACCURACY	PRECISION	RECALL	F1 SCORE
BERT	92.09%	92.75%	92.09%	92.16%
LSTM	89.09%	89.19%	89.09%	89.06%
SVM	88.95%	91.65%	88.95%	89.30%
Random Forest	88.81%	91.57%	88.81%	89.17%
KNN	88.54%	89.14%	88.54%	88.65%

COMPONENT CLASSIFICATION MATRICES				
MODEL	ACCURACY	PRECISION	RECALL	F1 SCORE
BERT	91.16%	91.16%	91.16%	90.09%
LSTM	90.07%	90.54%	90.07%	90.09%
SVM	86.68%	86.46%	86.44%	86.55%
Random Forest	86.44%	85.34%	86.68%	86.75%
KNN	85.91%	86.52%	85.91%	86.01%

Among the Sentiment Classification Matrices, BERT is the best model according to the highest score from all metrics: 92.09% accuracy, 92.75% precision, 92.09% recall, and 92.16% F1 score. These metrics each provide different insights into the performance of the model. The accuracy gives the ratio of correct predictions out of all the predictions made. This is a general measure for the overall performance of the model. Precision lets us know the percentage of the positive predictions that truly were, which focuses on how good the model is when it predicts a positive outcome. While recall is the ratio of actual positives that were returned as positive by the model-a fundamental value in determining how well the model identifies all relevant cases. F1 Score is the harmonic mean of precision and recall, providing a balanced measure of both false positives and false negatives. The fact that BERT surpasses

all these metrics means that it is especially good in picking up on sentiment and thus, technically speaking, is the most reliable model to handle the task at hand.

Next follows LSTM with good, solid marks: 89.09% of accuracy, 89.19% of precision, 89.09% of recall, and 89.06% of F1 score. This makes LSTM a tough competitor, too. However, the precision and recall of BERT are more important for sentiment analysis, since this relies on how well a model identifies positive and relevant cases without missing or misclassifying them. The remaining models include the support vector machine, Random Forest, and KNN, which have marginally lower performances, falling within an accuracy range from 88.54% to 88.95%. While this might indeed be the case, they are far less precise and lower in recall compared to BERT and LSTM, meaning there is a flaw in the correct identification of sentiment classes, possibly yielding more false positives or negatives in a real-world application.

Component Classification Matrices: Again, in Component Classification Matrices, BERT leads the chart with the highest values: 91.16% for accuracy, 91.16% for precision, 91.16% for recall, and 91.16% for F1 score. Component classification shall involve classification of various aspects of the supply chain into various classes of interest, such as retailers, suppliers, and warehouses. The meaning of the metrics here is similar: Accuracy gives the overall success rate, Precision ensures that whenever a model predicts a category, say a retailer, it is usually correct, Recall ensures that the model can find all instances of a particular category, and F1 Score balances the two aspects. It is further evidenced that BERT's performance has been consistent across all these metrics, giving it a better ability to distinguish between different components in the supply chain. This is what ultimately makes BERT so effective for this classification task.

LSTM also does quite well in component classification, with 90.07% accuracy, 90.54% precision, 90.07% recall, and 90.09% F1 score, hence being even more capable, though slightly less effective than BERT. The rest of the models, SVM, Random Forest, and KNN, are way behind in performance on all metrics. However, their accuracies range from 85.91% to 86.68%. Their F1 score is lower, revealing the fact that these models struggle more while effectively handling the various component categories and hence may lead to less reliable classification in a real-world supply chain scenario.

Conclusion

Taken into account, the results of both sentiment and component classification tasks are that BERT stands out as the best overall model. The obtained results are very consistent in that this model has the highest scores on accuracy, precision, recall, and F1 score in most cases, both on the task of sentiment and component classification, hence robust and having a higher generalization ability. Although LSTM also obtains high performance, especially on component classification, it turns out to be slightly inferior to BERT overall.

This means that when it comes to the general performance for both sentiment and component classification in supply chain data, the performance indicated for BERT presents

a strong model. It is also more accurate and balanced for all metrics, hence presenting the most reliable model for practical applications.

6. Conclusion and Recommendation

6.1. Blockchain

6.2. Sentiment Analysis

This research study has elaborated on the huge opportunities for incorporating blockchain technology into supply chain information-sharing systems and found that it could cure most of the ills of a conventional supply chain. With a decentralized, open, and unalterable ledger, blockchain holds the promise of bringing in disruptive changes in the way data is generated, stored, and disseminated to different agents along with a supply chain. The results of this study underpin the transformative power of blockchain to bring more transparency, efficiency, and trustworthiness amongst supply chain participants by motivating safer and seamless operations.

Among the important understanding derived from the research is the realisation that blockchain technology is not just an abstract prescription but actually has practical solutions which can be applied in various industrial segments. Through the literature review, empirical analysis, and case studies conducted in the course of this research, it is established that blockchain has been instrumental in solving a number of issues related to fragmentation of data, sharing of information in real time, and even data security. Ensuring all relevant stakeholders have the same correct and most current information limits the possibility of data manipulation risks, fraud cases, and inefficiencies attributable to traditional supply chain systems.

This study has also identified specific areas of the supply chain where blockchain integration will be most effective. It identifies industries that require traceability and verification, such as the food and pharmaceutical industries, as being better suited for blockchain integration. In these industries, blockchain can ensure the authenticity and safety of products, having a verifiable chain of custody from its origin to the final consumer. This means greater trust among consumers and better compliance with regulatory requirements by firms.

On the other hand, the study recognizes various challenges and limitations in the way blockchain is adopted into supply chains. In addition, there are also such technological limitations as scalability issues and the high computational resources that most blockchain-based applications require, in addition to full interoperability between different blockchain platforms. Moreover, the regulatory framework is still at the development stage in using blockchain technology and different levels of acceptance and regulatory oversight exist in various jurisdictions. These issues are indicative of the fact that although blockchain indeed looks very promisingly towards the future, careful planning and strategic implementation are what will be required in order to ensure that it meets all the expectations with regard to supply chain management.

This study develops a concept that provides a roadmap for businesspeople and policymakers interested in integrating the blockchain into supply chain operations. The framework delivered takes a step further with practical guidelines on how to overcome these identified challenges, underlines best practices for their successful implementation, and acts as a roadmap to lead the organizations on a path to realizing the potential of blockchains in building resilient, transparently efficient supply chains.

This research has contributed greatly to the understanding of the role of blockchain technology in supply chain management. It has given an overall analysis of how blockchain can solve the prevailing challenges concerning supply chains, provided practical insight with case studies, and even developed a conceptual framework for guiding future implementations. Results have shown that blockchain could be the use case game-changer for industries: solving pain points that have haunted industries for years and opening up new frontiers for innovation. While businesses continue to explore and adopt blockchain, additional research will be needed to further hone these strategies and unlock the complete benefit from this transformative technology.